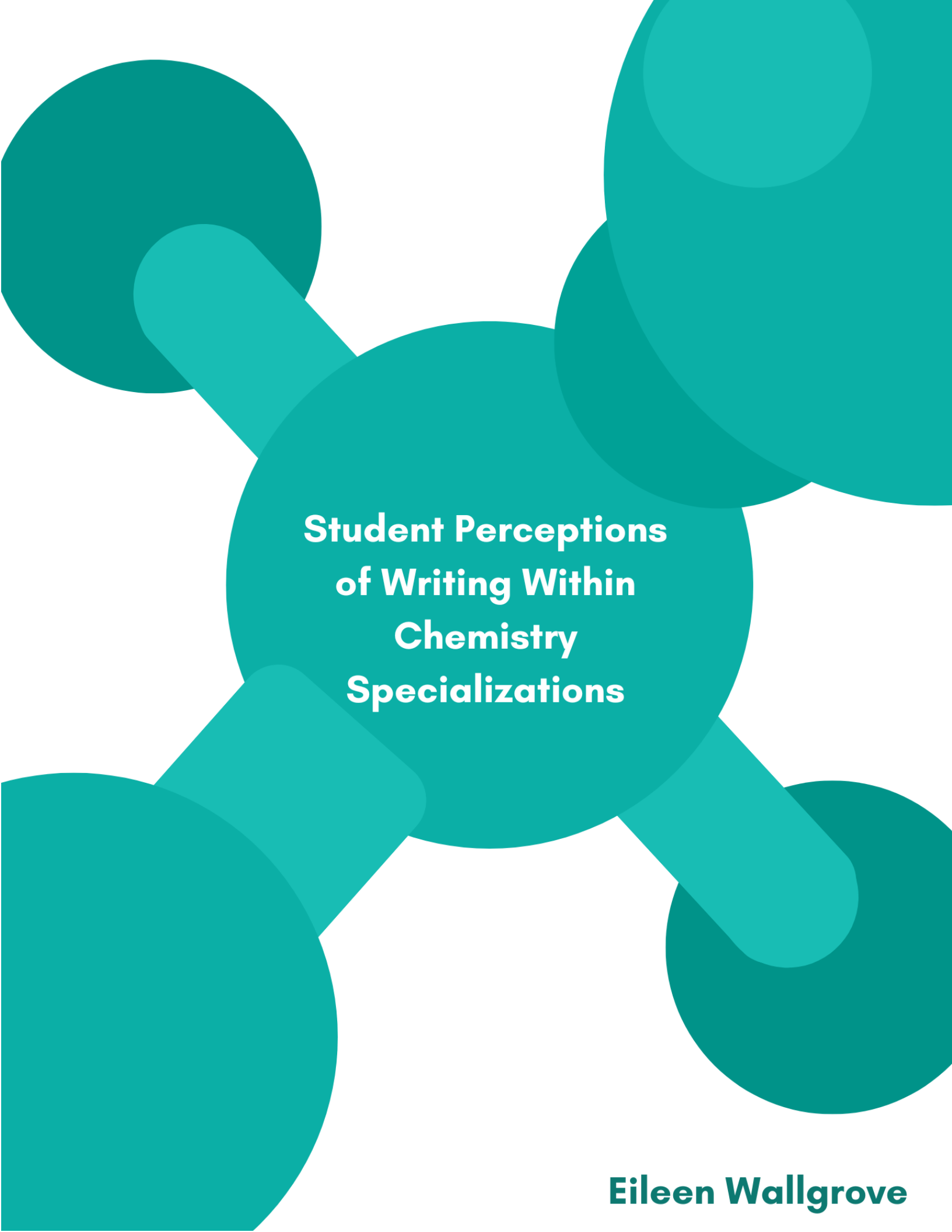


The background of the cover features an abstract design of overlapping circles and bars in two shades of teal. A central circle contains the title text. Four bars extend from this central circle towards the corners, each terminating in a circle. The circles and bars vary in size and opacity, creating a layered, geometric effect.

Student Perceptions of Writing Within Chemistry Specializations

Eileen Wallgrove

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Abbreviations

- **ACS** American Chemical Society
- **BREB** Behavioural Research Ethics Board
- **CSC** Canadian Society for Chemistry
- **CDQF** Canadian Degree Qualifications Framework
- **LTW** Learning-to-Write
- **REB** Research Ethics Board
- **SaP** Student as Partners
- **STEM** science, technology, engineering and math
- **U15** Group of 15 prominent Canadian Research Universities/Institutions
- **UBC** University of British Columbia
- **UERU** Association for Undergraduate Education at U.S. Research Universities, a group of US universities
- **WTL** Writing-to-Learn

Abstract

This exploratory study surveyed 149 students in chemistry specializations at UBC on their experiences and perceptions with writing and writing training (both generally and discipline-specific). Students were asked 54 questions covering demographics, experiences, perceptions, and suggestions regarding writing and writing training. Students were asked what courses they had taken, what writing activities and courses they felt belonged in a chemistry degree, and their perceived importance, enjoyment, and confidence levels regarding various writing-related activities and skills. Respondents reported a wide range of answers to most questions, demonstrating varied opinions. Collectively, three emergent trends were identified: a strong correlation between lab reports and students' perception of scientific writing, an exclusion of humanities-style courses, activities and skills, and a desire for more scientific writing training within their degree, particularly to address two lacking skills: oral communication and communicating to general audiences. The findings of this survey fill a gap of student perceptions within the literature, providing a student voice for future curriculum development at UBC and beyond.

Introduction

“Nothing in science has any value to society if it is not communicated” stated Ann Roe, in the *Makings of a Scientist* in 1953¹, demonstrating the universal importance of communication. Whether in academia, the private sector, or the government, professionals consistently rank communication skills highly, regarding them with the same or greater importance as technical knowledge.²⁻⁶ In chemistry, communication skills are leveraged when writing papers to share findings, discussing ideas amongst colleagues, or disseminating information to the public. The importance of strong, clear, and timely science communication was evidenced during the COVID-19 pandemic, where the world looked to scientists for advice on how to stay safe and limit risk.⁷ However, some scientists poorly communicated the scientific process and inherent uncertainty regarding an ever-evolving situation, resulting in mixed-messaging across different sources, prevalent misinformation, and widespread confusion.^{8,9} In this modern-world, scientists are now obligated to be masters of both technical and communication skills, needing to understand their discipline and communicate its essential findings or face the dire consequences.

The primary method of science communication is through writing, which encompasses several technical and contextual skills. Traditionally, writing skills are thought of as those regarding content and form such as word choice, grammar, syntax, and structure.¹⁰ These “hard skills” are typically taught to students throughout their grade school education in language classes, which is English for the purposes of this study.¹¹ More recently, academics have begun identifying other related skills as important to the science writer including audience identification and tailoring, ethical thinking, literary comprehension, database searching, and effective use of supporting imagery such as graphs, figures, and schemes.^{6,12-14} Thus, to be an effective communicator or writer requires a cohesive fusion of many skills as shown in **Figure 1**.

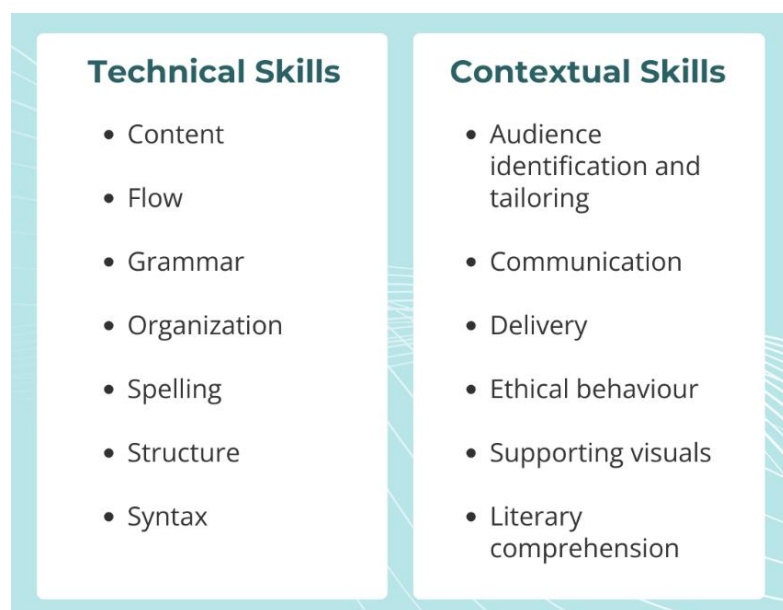


Figure 1. Technical (hard) and contextual (soft) communication skills

A further complication for new science writers – undergraduates, graduates, and post-graduates – is that science is very specialized; scientific writing reflects that, but their training lacks it. Many students' writing-training education has been in English classes without rigorous training in scientific conventions. The lack of scientific writing training may not have received such attention, had it not been well established that many graduate students^{15,16} and professional, publishing scientists are weak writers.¹⁷ The lack of training is global; in 2022, 28% of STEM graduate-students surveyed from 11 countries reported that they had had no training in an essential writing skill: 'using language appropriate for an audience'.¹⁴ 58% of those students had also not been taught the underlying theories of science communication and why it is important.¹⁴ **Appendix A, section 1** includes a tabulated reproduction of this survey results in **Table 18**. Without this training, undergraduate students have had to passively absorb discipline-specific practices including use of jargon (specialized language), citation method, structure, and form through exposure to the scientific literature. Only within post-graduate programs have some students been provided scientific writing training.⁴ However, this training is not even guaranteed in post-graduate programs, with only 16 to 34% of the same group of international graduate students reporting receiving training for the outlined communication skills within their graduate programs.¹⁴ Thus, it is understandable that without sufficient training, writing is a skill gap across the discipline.

The lack of accessibility for new science writers is not only an artifact of insufficient training, but a reflection of scientists' attitudes towards writing. During ancient philosophers, humanities and science co-existed in pursuit of knowledge and understanding. It was in the 1700s that progressive thinkers encouraged a division of the pursuit into the fields we now know as the social sciences (humanities) and physical sciences.¹⁸ From then, this divide solidified and became more nuanced with classifications of activities grouped under one or the other. This division caused scientists to develop an inaccessible practice of writing, tailored with specialized jargon known to those in the field, and creating in-groups.¹⁷ Recognition of the inaccessibility of scientific literature is documented as early as 1884.¹⁹ Yet the division remains, clearly evidenced by the separation of the Arts and Science faculties in most post-secondary institutions. A 2018 study performed by Moon et al.²⁰ surveyed over 5000 STEM faculty members of the Association for Undergraduate Education at Research Universities (UERU) and from those, a representative sample of 33 were interviewed in-depth. 40% of the teaching faculty reported not integrating any writing activities within their courses, with 15% of teaching faculty believing writing should not be taught within chemistry courses at all.²⁰ The resistance to incorporating writing training in science has led to insufficient writing training, implying to students that writing is peripheral in chemistry, perpetuating a detrimental cycle hindering interdisciplinary reconciliation.

In the early 21st century, a global standardization of post-secondary programs, including a renewed focus on increased scientific literacy and writing training occurred.²¹ The international reform of curriculum was started with the ratification of the Bologna Process standardizing educational practices across Europe.²² Canada followed suit in 2007, with a group of Ministers collectively publishing the Canadian Degree Qualifications Framework (CDQF) creating new degree standards for all Canadian universities.^{23,24} These degree standards remain in use, with the exceptions of Manitoba, Newfoundland and Quebec, who have their own, more stringent requirements.²² Alongside the standardization, at the academic level proponents started local integrations of new writing practices within the disciplines to improve both student writing ability through learning-to-

write (LTW) and student understanding of course content through writing-to-learn (WTL) initiatives.²⁵ While LTW is a traditional style of writing training, increased awareness of WTL has introduced writing into courses that traditionally did not have writing, offering students opportunities beyond writing training courses to practice writing skills in STEM degrees. Research papers, short-form summaries, poetry, creative assignments, podcasts, and peer review are some examples of writing activities that have been successfully integrated into chemistry courses.^{26–30}

The CDQF implemented broad degree curricula but didn't stipulate explicit guidelines to encourage significant writing training. The CDQF's guidelines around communication training pertaining not to any program but the entire bachelor's degree.^{23,24} The CDQF does not stipulate any requirements for breadth, and the only communication requirements are that programs should develop "the ability to communicate information, arguments, and analyses accurately and reliably, orally and in writing, to specialist and non-specialist audiences, using structured and coherent arguments, and, where appropriate, informed by key concepts and techniques of the discipline."^{23,24} The guidelines are intentionally broad to respect institutional autonomy, which "ensures that universities are free to pursue enquiry and disseminate knowledge based on evidence, truth and peer review."³¹ In respecting institutional autonomy but needing to ensure universities maintain a standard of education, the CDFQ stipulated that universities should offer an education comparable, or better than other institutions. Thus, it is the institutions which collectively decide on the implicit requirements for writing training.

The Canadian Society for Chemistry (CSC) and American Chemical Society (ACS) provide specific guidelines to chemistry programs across Canada and the USA through program accreditation. Regarding communication, the CSC also has slim requirements, as programs must only develop skills "including the writing of technical reports and presentations".³² The CSC does not state any specific learning outcomes but provides a few resources suggesting curriculum should be similar to other universities, meet provincial requirements, and their accreditation assessment. They do not explicitly state what they assess on.³² The CSC, unlike the CDFQ, mandates breadth with a minimum of 5 non-chemistry courses. However, inclusion of humanities courses is optional. The American Chemical Society (ACS) has more specific communication requirements for accredited programs. While the ACS only accredits American programs, the society is highly influential and sets out the following curricular requirements:

"writing and speaking opportunities that allow students to learn how to communicate technical information:

- (1) clearly and concisely,*
- (2) in a scientifically appropriate style for the intended audience including non-technical audiences,*
- (3) ethically and accurately, and*
- (4) utilizing relevant technology."*³³

The ACS also states that communication skills must be included and assessed in both written and oral form. None of the guidelines from the CDQF, CSC or ACS stipulate the extent of, or frequency for which, these skills should be developed and practiced; meaning, that for both the US and Canada there is no mandate for explicit writing training. Lab reports, which are discipline specific

and develop writing skills (and oral skills if orally presented), therefore satisfy *all* communication requirements presented by the three organizations.

Universities, with the power to determine their own curricula, have failed to implement homogenous writing and science communication training for their students. A preliminary analysis of the writing training at Canadian U15 institutes* (**Table 1**) showed that 40% of U15 institutions lacked *any* required communications or writing training courses as part of their physics, biology, and chemistry undergraduate programs. When looking at the offered scientific writing training, general or discipline specific, 67% of institutes offered scientific writing training courses but only 33% required students to take a scientific writing course as part of the BSc degree requirements. Chemistry-specific writing training courses were only available at four universities: UBC, McMaster, Montreal (in French) and Laval (in French), biology-specific at another four, and only McMaster offered a physics-specific writing training course. This variation in training both institutionally and interdepartmentally has resulted in no standardized writing or communication outcomes for BSc students. This deficiency was also noted regarding US and Australian curricula upon a review in 2015.³⁶

Compared to other U15 universities, UBC's writing training for chemistry specializations is more developed, but the experience is still inhomogeneous. Historically, to satisfy BSc communication requirements, UBC students took two writing training courses, one general and one discipline specific. For the general writing training course students chose between SCIE 113, WRDS 150, and a variety of ENGL 1XX courses. Whilst all communication courses, there were significant variations in content, activities and concerningly, learning outcomes between the courses. A student enrolled in SCIE 113 received scientific writing training, learning discipline-specific stylistic conventions and practices.³⁷ Whereas students in WRDS 150 and ENGL 1XX courses received a more generalized curriculum.³⁸ Further, even amongst the same course, e.g., WRDS 150, there were significant content variations between sections. WRDS 150 section "Do You Believe in Science? Science, Politics, and Beliefs" covered academic arguments for science, philosophy and politics as well as providing research and writing skills whereas section "Video Games and Diversity" discussed representation of minorities of gender, race, and sexuality within video game media.³⁹ The first-year general communication requirement could also be satisfied outside UBC with transfer credit from another institution or IB, AP or equivalent advance credit from high school. Thus, demonstrating the lack of standardized learning outcomes from the first communication requirement.

In third year, students satisfied the second communications requirement with CHEM 300. All chemistry specializations take CHEM 300, which provides students scientific writing training with a focus on chemistry content. Depending on their choice in the first year, CHEM 300 may be students' first exposure to scientific writing training. If so, this means many students completed at least half of their degree without any explicit scientific writing training. This not only harms students during those two years, who are left to passively determine discipline style-specificity but also leads to incoming CHEM 300 students having significant variations in experience and skill level, which is unfavorable for an upper year course. CHEM 300 does do a good job of, whilst having

* While there are more universities in Canada than the U15, the U15 were chosen as they host ~46% of all undergraduate students, and attain the majority of Canada's sponsored research funding.³⁴ However, we recognize that this sample does not represent all post-secondary institutions.³⁵

different curricula, graduating students with consistent learning outcomes and essential skills to be better communicators. While CHEM 300 standardizes learning outcomes for chemistry specializations, CHEM 300 is unique as it is one of only two major-specific writing training courses amongst the primary science disciplines (biology, chemistry, physics, math). The other option, SCIE 300, is restricted to students in a combined major in science. This again demonstrates the interdepartmental variation in writing training, suggesting that the presence of writing training is only dependent on whether the departments faculty have considered writing training to be an important addition. This is not ideal when science communication is an essential skill.

Table 1. Writing Training at U15 Universities, Current to 2023/2024 Academic Year

University	First-year English or French writing training	Biology-specific writing training	Chemistry-specific writing training	Physics-specific writing training	General science writing training
Alberta	Yes	None	None	None	None
UBC	Yes *	None.	CHEM 300 *	None	SCIE 113 ; * SCIE 300
Calgary	None	None	None.	None	Science 311 [∞]
Dalhousie	Yes	None.	None.	None.	SCIE 1111.03
Laval ×	Yes	BIO-3503 ; BIO-3502	COM-1920 ; COM-2920	None	N/A **
Manitoba	Yes	BIOL 3100 ^μ	None	None	None
McGill	None	BIOL 210 ; MIMM 301	None	None	WCOM314 ^β
McMaster	None ***	None.	CHEM 2Q03	PHYSICS 3D03 Inquiry in Physics	None
Montreal ×	None ***	BIO 4000	CHM 4000 ^μ	None.	ENG 1913 ^β ENG 1924 ^β ENG 1926 ^β ENG 1933 ^β
UOttawa	None	None	None	None	SCI 3101 ; SCI 4101
Queens	None ***	None	None	None.	None.
Saskatchewan	Yes	None	None	None	Can't find one.
Toronto ^α	None ***	None	None.	None	None ^α
Waterloo	ENGL 193 ; COMMST 193	None	None	None	COMMST 193 ; ENGL 193
Western	None ***	None	None	None	None
Summary:	7/15 (47%)	4/15 (27%)	4/15 (27%)	1/15 (7%)	7/15 (47%)

Bold are required courses

* 2024/2025-year SCIE 113 is required for all BSc students; chemistry majors take CHEM 300 instead of first year English

[∞] required only for chemistry and geology majors

× Indicates universities that are not CSC accredited

^μ only available to honours students

^β merged faculty of science and arts

^α offerings may vary by campus, table considers main campus; science writing training only at Trinity College (TRN478H)

*** Breadth requirements are present, writing/English courses could be included but not explicitly required

Perhaps, recognizing the inhomogeneity, the UBC Faculty of Science has changed the general communications requirement for the 2024/2025 academic year. All incoming students will now be required to take SCIE 113 as their first-year writing training course.⁴⁰ This should significantly standardize the writing training across the science faculty, including amongst chemistry specializations, ensuring all science majors will have scientific writing training within their BSc degree. However, will this be enough to make students effective science communicators? To broach this topic, we ask the research question: “What do students think about writing and writing training within chemistry?”

Student perceptions of writing are influenced by academic, social, and personal factors. (**Figure 2**) The initial exposure most students have with writing is within grade school, using writing to convey their learning to their teachers on the journey to literacy.⁴¹ In these formative years of development, students are exposed to the systemic beliefs about the roles of writing and humanities within the sciences from sources at their school and outside, shaping their initial perceptions about what writing is and where it is relevant. Student perceptions will be further shaped as they grow by their personal experiences with writing outside the classroom – for personal or career reasons. Once reaching post-secondary education, they will enter a secondary academic environment with a different environment and integration of writing. We would expect students opinions to mirror their main writing influences, such as teaching faculty.⁴² However, there is little literature documenting what students think should be present in their chemistry degree curriculum and even less on student perceptions of writing within chemistry.

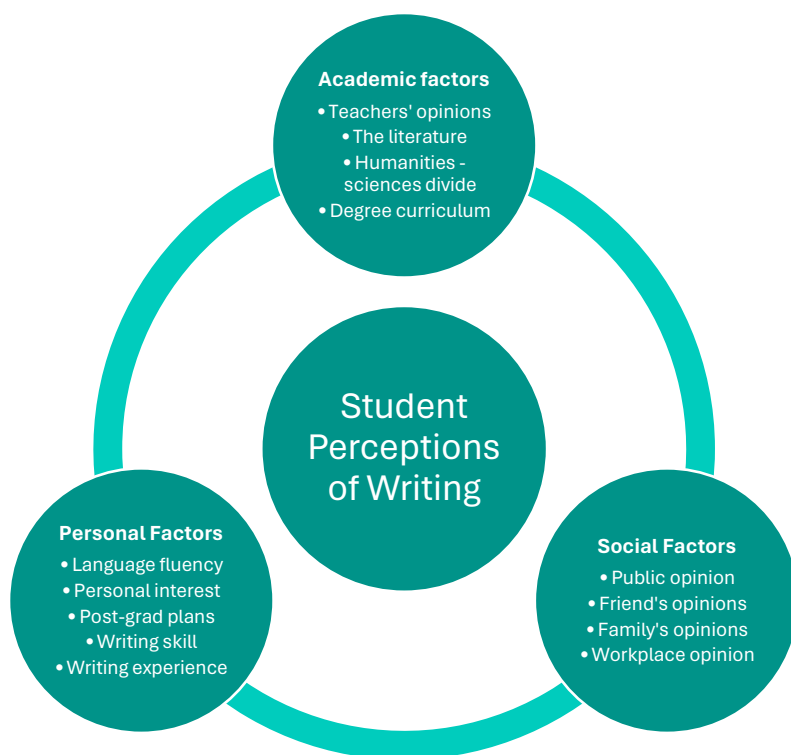


Figure 2. List of academic, social, and personal factors influencing student perceptions of writing

Students' complaints about a lack of voice in their curricula development are well documented, and over the last decade have started to be addressed by institutional efforts under the students as partners (SaP) initiative.^{43,44} These initiatives address students' desires to be active partners in their degree and curricula development, rather than just passive learners.⁴³ In these partnerships, select students provide feedback on curriculum development, ensuring the learning outcomes they need taught are included. However, as this is an initiative aimed at faculty, most data collected from these initiatives remains stored within institutions, is often hyper-specific to certain courses, and often represents only the opinions of a few students. Therefore, curricula are still predominantly developed by the faculty. While the experience and knowledge of student learning methodologies possessed by science educators is substantial, it is not a replacement for the student voice. Students are active within their education, and we cannot discredit their lived experiences and opinions on their own education.⁴⁵

Large scale data collection on students' opinions are not generally performed beyond Student Experience of Instruction surveys (SEIs) and brief alumni surveys. At UBC, SEIs are completed at the end of each semester and collect student feedback on the courses they've taken.⁴⁶ These surveys are primarily used to gauge the instructor's performance (tied to the instructors' performance reviews and career progression) rather than a review of the course content. Alumni surveys, while helpful in assessing an overall degree, must be brief to gather key insights in a population that has little motivation to answer the survey and often have low response rates. The BC Student Outcomes alumni survey, for example, collects only some basic demographics information, employment information, and asks students how well the program developed certain skills.⁴⁷ For former UBCV chemistry students, (**Appendix B**) 79% agreed that their degree was helpful or very helpful in developing the skill of writing clearly and concisely, compared to the humanities, in which 91% of students agreed. While this information is useful in addressing satisfaction, the university's primary motivation, it doesn't tell us why over 20% of students didn't agree that their degree was helpful in developing writing skills or what they think of writing within chemistry degrees overall.

The lack of student voice on writing isn't unique to UBC, there is a distinct lack of publications that consider students' self-reported perceptions regarding writing and writing training within their undergraduate program. The greatest volume of work regarding students' self-reported perceptions of writing within the sciences originates from several authors in Australia. Varsavsky et al. surveyed students in their final semester of a three year BSc.⁴⁸ The survey asked students to self-report on core skills as developed in the Science Students Skills Inventory (SSSI).⁴⁹ Students were asked to rank on a four-point Likert scale from not at all to very on six indicators regarding communication skills: importance, assessment, inclusion within program, improvement, confidence, and future use. On average, students ranked the importance of writing highly (3.70) with 96.3% of all students ranking it as important or very important. Students also ranked the importance of writing higher than their improvement of the skill during their degree (3.36).⁴⁸ Mercer-Mapstone and Matthews (2018) performed a very similar study using the same four-point Likert scale and questions, surveying 2566 Australian BSc students in first, second and third year.⁵⁰ Students here, also felt more strongly on the importance of writing skills than how much their writing skills improved (3.51 vs 3.01). Students' confidence levels regarding writing were 2.98 and 3.24 respectively, suggesting that the perceived lack of improvement in writing was not due to student's overestimating their

skills.^{48,51} Thus, both studies show that students identified writing as an important skill but did not receive the same volume of training relative to the perceived importance.

This study aims to align with current movements recognizing the essentiality of writing and increasing the volume of writing training within undergraduate curriculum, by presenting the student voice. In a novel study of scope and population, we assess student perceptions on not just writing, but also specific writing activities and skills commonly encountered within chemistry degrees. This expands on the existing literature which has, as far as we know, only considered students' opinions on writing broadly or in the context of very specific writing courses or activities. This is also, as far as know, the first presentation of findings regarding *Canadian* student opinions of writing within their undergraduate science degree. We hope that this presentation of the student voice provides valuable insights into successful endeavors but also areas of improvement to inform future curricula development efforts both at UBC and in broader educational contexts.

Methods

Ethics Approval

All research involving human participants is required to receive institutional ethics approval before commencing.⁵² Ethics approval is a legally mandated process that evaluates a study on the level of risk to participants, risk-mitigation efforts, data management practices, and collection of informed consent.⁵³ Proper informed consent includes detailing the survey practices, risk, and outcomes to participants and collecting their agreement to the survey in an immutable record.⁵³ The process of ethics approval is essential in both upholding human rights and maintaining the public's voluntary participation in research by ensuring participants feel safe and informed throughout the research process.⁵² This study was performed at UBC, so the ethics approval was conducted by the UBC Behavioural Research Ethics Board (BREB).⁵⁴

The UBC BREB ethics approval process involved three stages: application scoping and drafting, submission, and review. The most time-consuming process was the first stage as it involved nearly all preparatory work required before the study was launched. For this study, stage 1 involved defining the survey goals and scope, participant demographics, and drafting the survey questions. Other concerns pertinent to the BREB were compliance with proper data storage procedures, thorough information regarding how informed consent was to be attained, and other survey activities including advertising and the follow-up focus groups. (The focus groups have not yet been held at the time of writing this paper). Stage 2 entailed filling out the ethics application through the online UBC Research Information Systems (RISe) platform.⁵⁵ Properly formatted supporting documents including the survey questions, consent form, email communications, and advertising materials also had to be attached. (See **Appendix C** for the approved ethics application excluding attachments.) Stage 3 involved the revision of the ethics application addressing the BREB's Provisos – requested corrections and clarifications – and submitting an updated application that addressed their concerns. The requested Provisos by the BREB committee were minor: a few

formatting changes and minor content changes to the attached documents to ensure each document contained all necessary information. No large changes in survey content or study practices were required, and ethics approval (H23-03924) was granted by the BREB on January 18th, 2024.

Survey Design

The formation of this study involved an extensive literature review culminating in a survey to address the research question “What do students think about writing and writing training?”. From this question, five survey themes were identified:

- 1) Students’ understanding of how writing training fits into their chemistry degree, and the role of writing and writing training within chemistry at large
- 2) Students’ writing training wants, needs, and satisfaction within their chemistry degree
- 3) Students’ relationship with writing and writing training over the course of their degree
- 4) Impact of different backgrounds and experiences on student perceptions of writing and writing training
- 5) Suggested areas and modes of improvement regarding writing training at UBC

These themes originated from the personal interests of the research team (as science communication educators) and literature. The conclusions from the literature review were covered within the introduction, with the key takeaways that there is a lack of literature regarding student opinion worldwide and as far as we know, no literature on students’ perceptions on writing from within Canada. Thus, a survey of exploratory nature was deemed appropriate to cover a broad range of topics and questions to probe for interesting or unusual trends among students to be further explored in future works.

The survey was 54 questions long, comprising four main sections: demographics, student experiences, student perceptions, and several additional questions. The goals of each section and question contents are listed in **Table 2**. A full copy of the survey and all survey questions is in **Appendix D**. Each section of the survey directly addressed two or more of the five survey themes while influencing the others. Demographics addressed themes 3 and 4, to see how student background and academic year influenced student perceptions. Student experiences addressed all themes by informing what writing assignments and training students based their opinions of off. Student perceptions addressed themes 1 and 2 with questions probing what students understand as the role of writing in chemistry and their degree. Finally, the additional, free-answer questions addressed themes 2 and 5 by allowing students to respond with what writing and writing training was useful and what areas they still need more support in. Questions not part of these four sections were also included in the survey for participant consent and to facilitate the prize draw and focus group recruitment.

Table 2. Summary of the Four Survey Sections

Section	Goal	What we asked: Questions / Content
(1) Demographics	Explore how students unique, personal background impacts their opinions	Academic year Age range Co-op experience Gender HS education – region and language of instruction Primary language Specialization
(2) Student Experiences	Explore what writing and writing training experience students have at UBC	Writing related tools and resources used Writing training courses taken Writing assignments completed within degree
(3) Student Perceptions	Explore what students think about writing and writing training at UBC, if it's necessary, sufficient, useful, important, and what role writing has within chemistry at large	What writing skills/activities are: - Important within a chemistry degree - Important for students' future goals - Enjoyable Students' confidence in their writing ability If a chemistry degree should contain <i>general</i> writing training If a chemistry degree should contain <i>scientific</i> writing training What types of writing courses should be in a chemistry degree What types of writing activities should be in a chemistry degree
(4) Free answer	Let students freely answer if they need more writing training and what things they like about the current writing and writing training at UBC	What areas need more <i>general</i> writing training What areas need more <i>scientific</i> writing training What writing assignments were most <i>useful</i> and why What writing assignments were most <i>enjoyable</i> and why

The survey design was intentional, placing important questions earlier and structuring question answers to avoid bias. Content questions regarding student experiences and perceptions were considered most crucial and placed first to increase response rate,⁵⁶ followed by demographics and then optional, free-answer questions. Demographics was placed near the end to avoid biasing student responses^{57,58}, which was preferred over potential gains in completion rate from an earlier placement.⁵⁹ Survey questions with multiple answers were predominantly ordered alphabetically to avoid bias.⁶⁰ Exceptions were made when there was an inherently logical ordering to the answers such as increments of time or amount, or to put an “other”, not applicable, or “none” style-answer at the bottom in standard survey convention.⁶⁰ The Likert scale questions were structured in ascending order (SD to SA) to correct for ‘left side bias’ which results in overinflation when Likert questions are in descending order.⁶¹ We also chose to include a neutral option “neither agree nor disagree” for Likert questions that asked about agreement. Neutral options are a valid sentiment of survey participants and exclusion of a neutral option can overestimate participant’s sentiments.⁶²

Various styles of survey questions were used to create a clear and efficient survey to maximize participation rate. Questions that are fast to answer, i.e., multiple choice and Likert scale, comprised most of the questions, with a few short answer follow-ups, a few yes / no questions, and four long answer questions in the optional section. Long answer questions were kept to a minimum to avoid participants from dropping out part way and maintain a reasonable survey length.⁵⁷ A progress bar was not included due to recommendations from Qualtrics (the survey platform) and evidence from the literature that a progress bar decreases survey completion rate.⁵⁶ Even with these efforts, the survey was longer than recommended by the platform and the literature⁵⁶, which

is why all questions were made optional, so students could skip questions they didn't want to answer.

Certain questions were inspired from the literature to offer some comparison value. One of the most prominent set of survey questions present regarding student perceptions within the sciences are those based on the SSSI.^{12,48,49,63} The SSSI, while built relative to Australian post-secondary curricula, consider skills that are relative to science students worldwide. The skills included in the SSSI are scientific content knowledge, writing skills, communication skills, teamwork skills and quantitative skills. The first use of the SSSI in a survey asked students to rank on a Likert scale of 1 to 4, the skills on five indicators: how important, their level of improvement, the level of inclusion of these skills within their degree, and their confidence using these skills. In this study, we also chose to ask students to evaluate certain writing related skills on the metrics of importance, confidence, and a new factor, enjoyability.

Questions pertaining to demographics and certain chemistry skills were also sourced from the literature. Kaur and Prendergast's insightful discussion on the implications of English as a secondary language encouraged our demographics questions that considered students primary language.⁶⁴ Skills including literature databases, word processors, and ChemDraw / graphing/ visualization software were sourced from a study by Kadtsyna et al.¹³ Kadtsyna et al. also inspired the question which asked if writing was more useful for certain chemistry subdisciplines, based off questions that asked if certain skills were more useful for certain classes or courses.¹³ The question regarding whether outcomes of students work such as in a career setting impact their motivation was modelled off Richardson et al.'s work that suggested students were more motivated to write lab reports when they served a purpose, in their case being used by other students.⁶⁵ Questions regarding whether students found writing helpful to clarify misconceptions were inspired by the book "Reaching Students" which found that implementing WTL forced critical reflection on the students, increasing learning gains.⁶⁶

To increase survey participation, participants were entered into a random draw for one of fifteen \$10 Amazon gift cards. The random draw to select the winners took place after the survey ended and was performed using a random number generator within Excel. Incentives were emailed out as electronic gift-cards to the winners. As the incentive was minimal, it was not expected to have caused any bias in the respondents.⁶⁷ To comply with BREB regulations, participants did not have to complete the survey, only submit their contact details to enter the draw.

All students were informed that the survey was confidential, with student's personal information gathered solely for the random draw or, for students who indicated interest, the follow-up focus groups. Students' incorporation of their personal information should not have influenced results as students were clearly informed that the survey results would be kept confidential, and their personal information would be de-identified before being shared with the rest of the survey team. All data was stored in FIPPA-compliant Canadian servers hosted by UBC.

Pilot-test, data collection and analysis

After the survey was drafted, a pilot test was conducted and survey feedback was incorporated. For the pilot survey, we collected feedback from a varied group of undergraduate and graduate

students and faculty members of both the teaching and research streams in chemistry (N = 7). Feedback was collected in a Word Document form that asked pilot testers how long it took to complete the survey, if any questions felt unnecessary or confusing, if there were any missing ideas or skills, and to list any other improvements they had. Feedback regarding survey content was varied, but two items of feedback were common: (1) that the survey should be reorganised and (2) that the survey was too long. To address the first point of feedback, we reorganized the survey into its current structure, including moving demographics to the end. To address the second point of feedback, we made the personal perceptions and long answer questions optional at the end of the survey and cut out a few less essential questions. Other changes included fixing grammatical errors and adding “Data / Molecular Visualization Software (e.g., ChemDraw, Tableau, Graphing Software)” as a writing related tool to recognize the importance of supporting imagery in writing.

The survey was then shared with all students specializing in chemistry or chemical biology or minoring in chemistry. Students enrolled in biochemistry were not invited to participate in the survey as they primarily take biochemistry instead of chemistry classes, however the responses from a few who did participate were considered. As the first year students had not yet declared a major when the survey was distributed, all students enrolled in CHEM 123 were invited if they were considering a chemistry or chemistry related specialization. Any response from first-year students who indicated interest in pursuing majors outside chemistry, chemical biology, or biochemistry were not considered.

The survey was emailed to all in-scope participants on February 16, and left open for three weeks, until March 8th. Participants were invited to complete the survey via an email delivered from the undergraduate program coordinator, posters within the UBC Chemistry Building, and in-class advertisements. A reminder email was sent on March 4th, that also extended the deadline. The survey was extended to increase the response rate, as the survey was released over reading week and the following week was when most midterm exams were occurring. Thus, we hoped that an extension and reminder would prompt students to complete the survey who were previously too busy.

The survey results were tabulated and statistically analysed by methods recommended within the literature. Proper analysis of Likert-scale questions is of ongoing debate^{68–72}, and as the normality of the data varied, both parametric and nonparametric analyses of variance (ANOVA) tests were performed. The parametric one-way ANOVA test was performed within Excel with subsequent T-tests for any statistically significant p-values. For this study, a standard confidence interval of 95% was chosen. F-Tests were employed to test for variance when ANOVA tests were not performed. The nonparametric ANOVA was a Kruskal Wallis H-Test followed by a post-hoc Mann Whitney U. Results of both ANOVA tests were compared and while most results agreed, some differed. In these cases, the normality of the data was examined to determine the superior test.

Multiple choice questions were broken down into individual answers, counted and analyzed using a chi-squared test of independence.^{14,73–75} A chi-squared test was deemed appropriate as while the variables are options from the same question they are not competing (i.e., the question asked students to select *all* activities not just a subset such as the main activities).⁷⁵ The Kruskal Wallis and chi-squared tests were performed using an online calculator from Statistics Kingdom.^{76,77} While Statistics Kingdom is not accredited, it is an open use resource with calculations based in

'R'. A comparison using the Statistics Kingdom parametric ANOVA calculator and Excel yielded the same results, indicating the calculations are valid.⁷⁸ However, it remains a potential source of error.

Long-answer questions were analysed using a thematic analysis loosely following the methods suggested by Bree and Gallagher.⁷⁹ After a preliminary examination of the answers, repeating sentiments were identified and chosen as topics. All answers which contained that topic were marked on an Excel sheet and the counts were summed for a complete tabulation. Certain themes were highlighted in corresponding colours within the answers. Questions regarding what activities were most useful or enjoyable were divided into three categories of topics: location (i.e., what courses), activity (e.g., essays, summaries, lab reports) and reasoning (i.e., why it was enjoyable or useful). An open-response question was included at the end allowing students to provide any additional feedback to the department. A thematic analysis was performed on these answers; however, they were too varied to be included within this report.

Results

Section 1: Demographics

The survey received 367 responses, including incomplete responses and responses from participants outside the survey scope. After filtering for participants that consented for their data to be used, responses that passed the attention check, and chemistry related specializations (chemistry, chemical biology, and biochemistry), **Figure 3A**, there were 149 responses. Most participants were first year undergraduate students ($n = 93$), with 19 second year students, 21 third year students, and 16 students in fourth year and beyond. This is shown in **Figure 3B**. The survey represents a small proportion of the approximately 275 students enrolled in chemistry.⁸⁰ All data pertaining to Chemistry specialization program demographics was obtained from PAIR as of 2022/2023.⁸⁰ While the data is from the previous academic year, demographic trends in chemistry are relatively consistent year-over-year, suggesting this data is a good approximation.

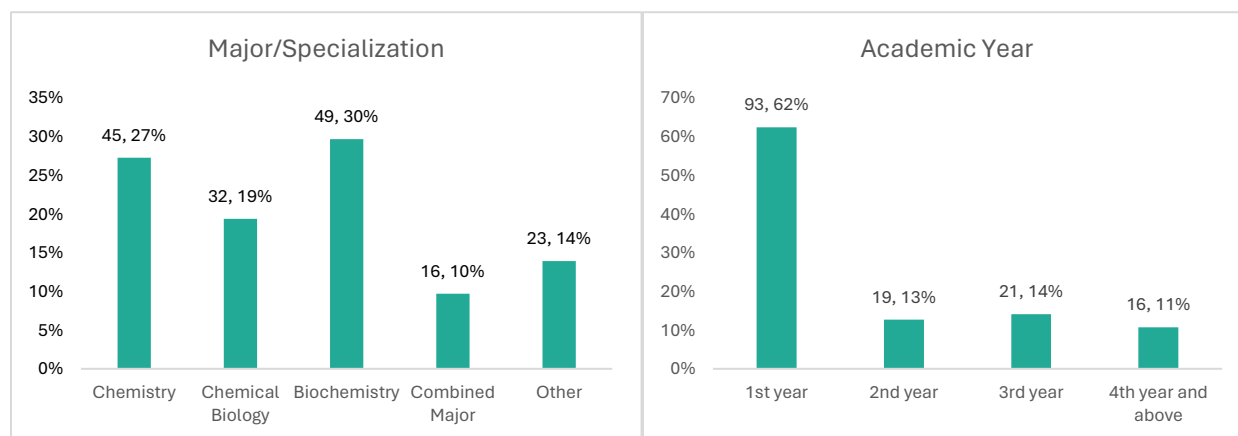


Figure 3. (Left, A) Responses to “What is your specialization (or intended specialization)?” ($N = 149$) and (Right, B) “What is your current academic year (excluding co-op work terms)?” ($N = 149$)

Students were also asked for personal information, including age and their preferred gender identity. Rather than ask for specific ages, we asked if students self-identified as mature students; that is, over 25. All students reported as being under 25 in line with the estimated student age of 21.1 across the chemistry program.⁸⁰ Most survey respondents identified as female (n = 109, 73%) with 33 students identifying as male (22%), 6 as non-binary (4%), and 1 selection of unsure/prefer not to answer (**Figure 4**). Non-binary and unsure/prefer not to answer were considered together as a gender minority to allow for meaningful analysis. Compared to the entire chemistry program where 57% of students are female and 43% are male (n.b. unmentioned), female participants are overrepresented within this study.⁸⁰

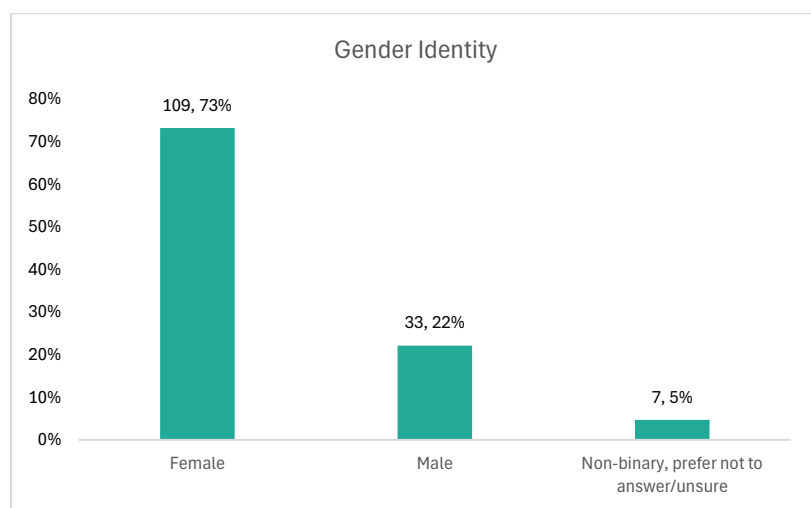


Figure 4. Responses to “Indicate your gender identity:” (N = 149)

As fluency deeply impacts writing ability, several questions were posed regarding student’s primary language and education background. Most students (n = 126, 85%) identified English as a primary language, their high school education as being primarily in English (n = 137, 92%), and their high school education being within Canada (n = 128, 85%). The associated graphs are in **Appendix A, Section 2**. Program wide statistics show that 75% of chemistry students are domestic students.⁸⁰ The survey results may appear to suggest that most respondents are native English speakers from Canada and international students are underrepresented within the sample population. However, international students can complete their high school education within Canada or at a school abroad that teaches in English, which could be obscured within how these questions are phrased.

Students were also asked if they had completed or were in the process of completing a co-op placement. We theorized exposure to work environments may offer new insights to students influencing their perceptions of writing. Only 14 students (9%) indicated having been or being enrolled within co-op. It is also important to mention that as co-op is only available starting in third year, co-op students are an explicit subset of third- and fourth-year students. Despite the small number, we still analysed based on co-op experience.

The last demographics question we asked students was their post-graduate plans (**Figure 5**). Most students intended to pursue graduate school (n = 66, 44%), or a professional program such as accounting, dentistry, law, or medical school (n = 39, 26%). Following that, a significant number of students answered that they were undecided (n = 27, 18%) and a few answered they wished to enter the workforce (n = 17, 11%). The survey respondents appear to have a strong sense of direction after their degree, which is surprising given that most participants were first-year students. The preference for graduate school is expected, as most research chemist jobs require a postgraduate degree.⁸¹ Students may also be interested in pursuing graduate studies for the research involved in the theses.

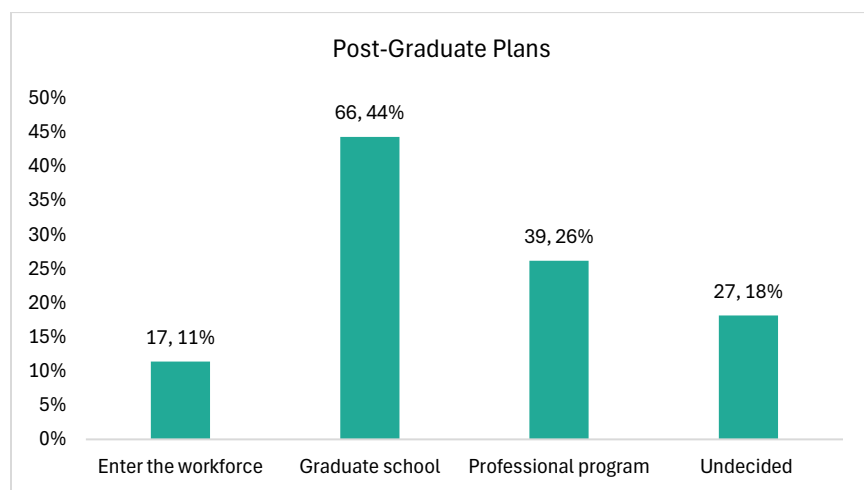


Figure 5. Responses to "What are your plans after graduation?" (N = 149)

Section 2: Student Experiences

The second portion of the survey asked students about their experiences with writing and writing training at UBC. The aim of this section is to provide a brief overview of those questions as context for Section 3: Student Perceptions. This section explores and attempts to quantify what writing training courses, activities and resources students have had or used within their degree. Certain questions that were not discussed are presented within **Appendix A, Section 3**.

Students were first asked what writing training courses they took and when. As expected from the inhomogeneous course offerings, students completed a wide variety of writing training courses as shown in **Figure 6**. The most enrolled courses were first-year writing training courses including ENGL 1XX (n = 55, 37%), WRDS 150 (n = 45, 30%) and SCIE 113 (n = 45, 30%). Approximately equal proportions of students had taken one of the three courses, with the SCIE 113 participants being predominantly first year students (n = 36, 80%). This distribution is expected given the high number of first-year respondents coupled with the recent shift towards preparing SCIE 113 to be a mandatory course. All but two upper year students i.e., years 3 and 4, had taken CHEM 300 (n = 35, 23%). A few students (n = 7, 3%) selected CHEM 300 without any first year-English courses, suggesting some students may have misremembered or have had alternate credit. Twenty-three

students (15%) had not taken any writing training courses at UBC. Most of these students were first years, with two second years and one third year. None of the students were transfers, suggesting students either had alternate credit or forgot they took a writing course. While we did not specifically ask regarding writing training outside UBC, some students did specify it in the “other” category. Most students had taken one of their courses within the current academic year (n = 88, 59%) with a smaller number of students having taken it in the previous academic year (n = 33, 22%). Very few students had taken it two or more years ago (n = 5, 3%). (**Figure 7**) Overall, it appears that most students correctly identified their prior writing training and had recent writing training experience. This suggests that students can effectively recall these experiences and provide relevant, accurate opinions regarding current writing activities and training at UBC.

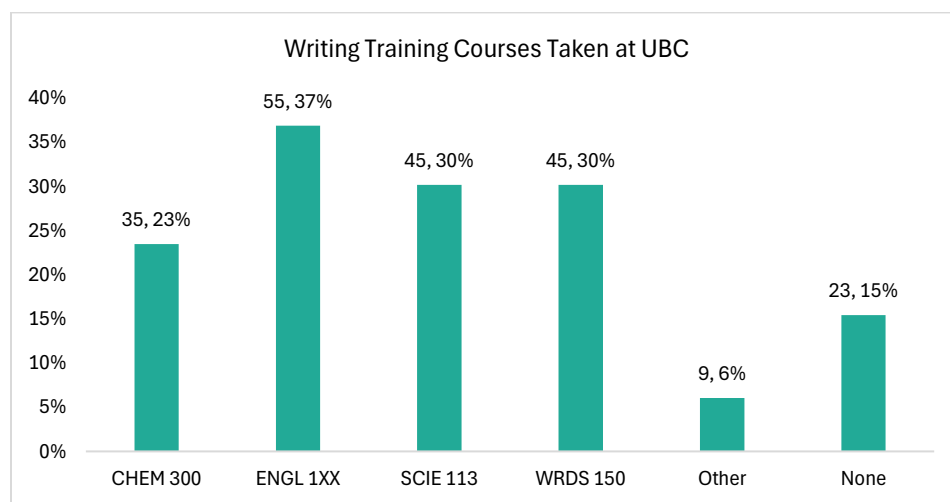


Figure 6. Responses to “Which of the following writing training courses have you taken at UBC? Select all.” (N = 149) “I do not remember what I took” and “SCIE 300” were also answers but not chosen (n = 0).

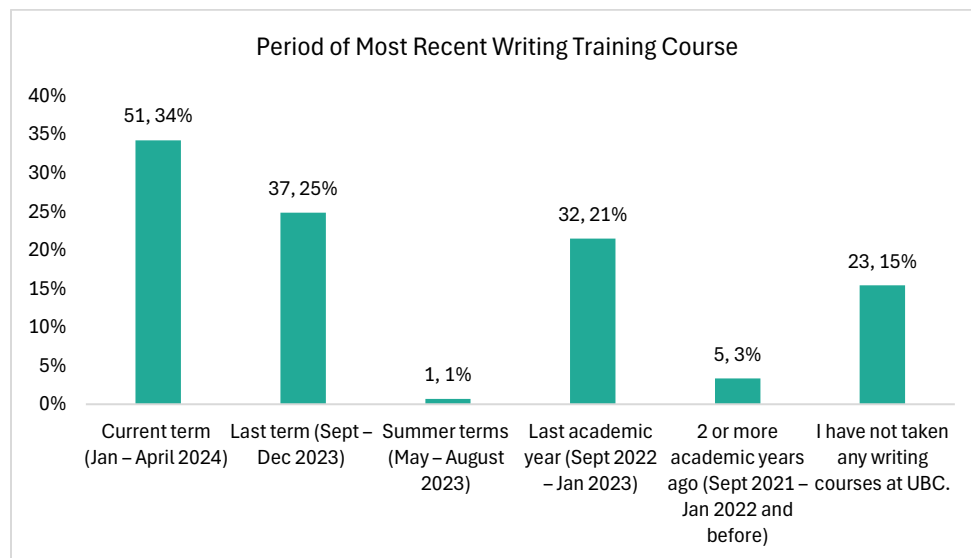


Figure 7. “When did you take your most recent writing training course at UBC?” (N = 149)

Students were also asked about the content of writing within their chemistry courses. 70 students, (47%) answered that at least one of their courses had a significant amount of writing and specified which courses as presented in **Figure 8**. The most common responses were CHEM 300 (n = 30, 43%) and first year labs i.e., CHEM 121/123/141 (n = 23, 33%). Students included a range of other courses, but the most prevalent theme was that students felt all lab courses i.e., CHEM 121/123/141, CHEM 203, CHEM 245, CHEM 3XX, and CHEM 44X had a significant amount of writing. Some lecture courses were identified, e.g., CHEM 218 and CHEM 327, however, in a lesser number. For what types of activities students identified as being present in these courses, lab reports were the most common choice (n = 59, 84%) followed by reports or research papers (n = 32, 46%), presentations (n = 26, 37%), and then the others (**Figure 9**).

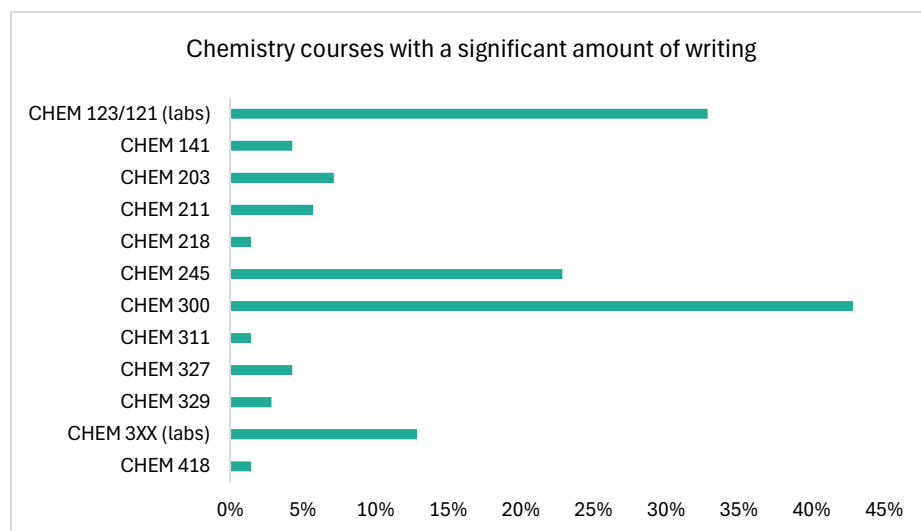


Figure 8. “What chemistry courses have you taken that you think had a significant amount of writing assignments or activities?” (N = 70, count = 103)

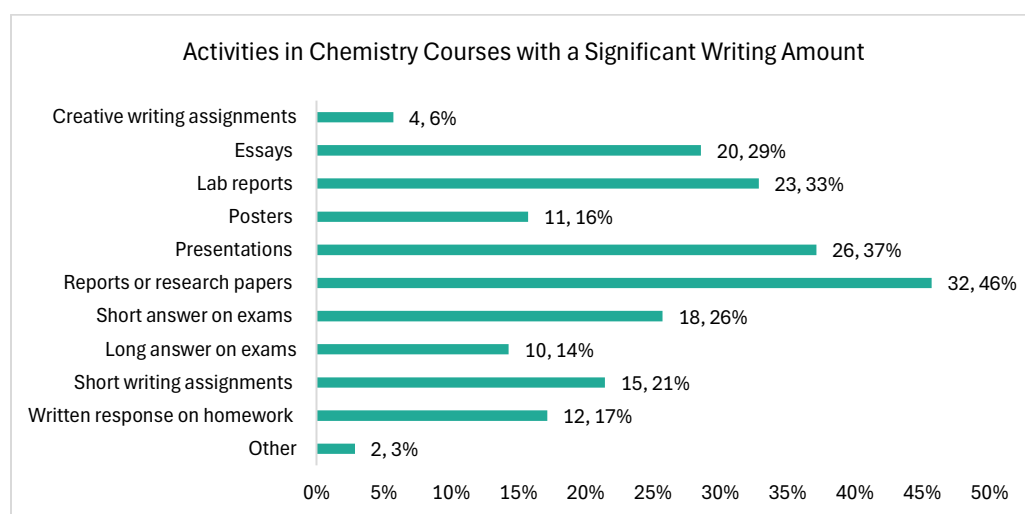


Figure 9. Responses to “What types of writing assignments or activities were in those chemistry courses with a significant amount of writing? Select all.” (N = 70, count = 173)

Students were also asked if they used certain writing related tools, methods, and software. **Figure 10** presents the different tools and university provided resources such as the UBC library or UBC Centre for Writing and Scholarly Communication (UBC CWSC). Students were instructed to select all they had used within their degree, and most students (n = 146, 98%) had used at least one. The UBC library was the most used resource (n = 122, 82%) followed by word processors (n = 114, 77%). The usage rate of word processing software seems low, given that we expect most students to submit typed writing assignments and that all students have access to Word via UBC's free Microsoft Office student license.⁸² This suggests that students may not know what a word processor is and may have not recognised experience with other common programs like Google Docs as applicable to the question.

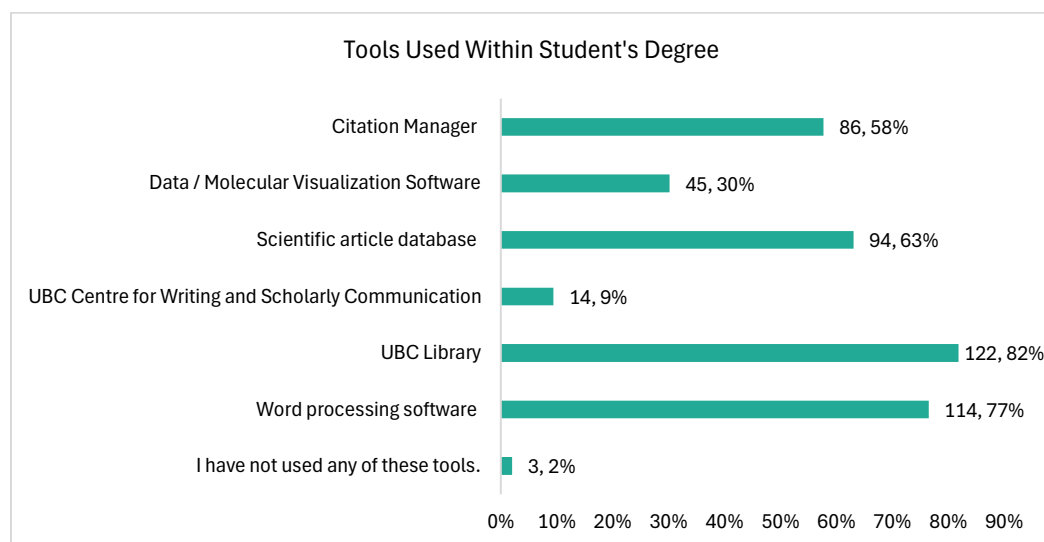


Figure 10. Responses to “Which of the following writing related tools, programs, and websites have you used in your degree? Please select all.” (N = 149)

Students were then asked how frequently they used a similar set of tools/software when writing a final project, essay, or research paper. Frequency of use varied between the different tools as shown in **Figure 11**. Students were using multiple drafts (4.28 ± 1.47) and outlines (4.01 ± 1.43) most frequently with 46% of students also using peer review at least 50% of the time, suggesting students collaborate on many assignments. Not many students used a CWSC consultation despite it being free and anonymous suggesting a barrier to use, such as low awareness, fear, or incompatibility with student's writing processes. Generative AI was the least used tool, with only 13 students (9%) using it 50% of the time. This low usage rate could be from unfamiliarity or ethical and accuracy concerns. Students may have also been cautious of admitting generative AI use due to academic misconduct concerns, even though the survey was confidential. Social desirability bias may have also encouraged students to want to display ‘good behaviour’ by not selecting generative AI as a tool commonly used to the university affiliated survey team. More students answered using an online proofreading software such as Grammarly 50% or more of the time (n = 98, 66%) which is also AI however, more accepted as a tool rather than a writer replacement.

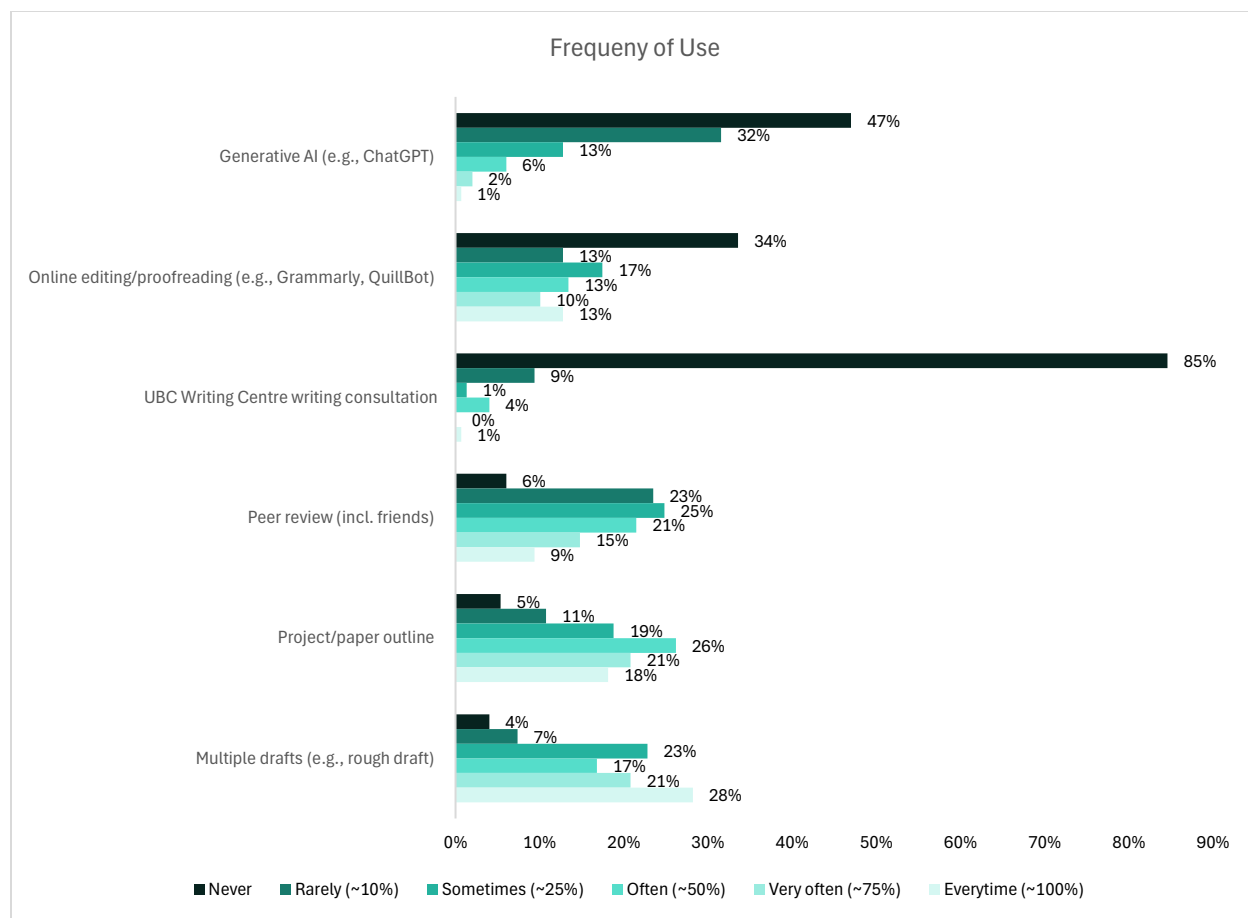


Figure 11. Responses to "When writing an essay, final project, or research paper assignment, how frequently do you use the following methods:" (N = 149)

Section 3: Student Perceptions

Section 3 considered students' feelings and perceptions around writing and writing training. Students were asked where, when, and how writing should be integrated into a chemistry degree, if at all. Students were also asked about the importance and enjoyability of various skills and activities and their perceived skill level as a writer.

When asked about the amount of writing, most students agreed that the current amount of writing training was sufficient. **Figure 12** shows that 102 students (68%) answered that in their degree and to improve their ability as a writer there should be the same amount of writing. There was a slight preference amongst students for more or a lot more writing (n = 29, 20%) as compared to less or a lot less writing (n = 18, 12%). **Table 3** shows the average values with the scale from 'a lot less writing' to 'a lot more writing' codified from 1 to 5. There were no statistically significant differences ($p < 0.05$) in the amount of writing training by gender, academic year, major, co-op experience, primary language, whether students had had writing training at UBC, or post-graduate plans.

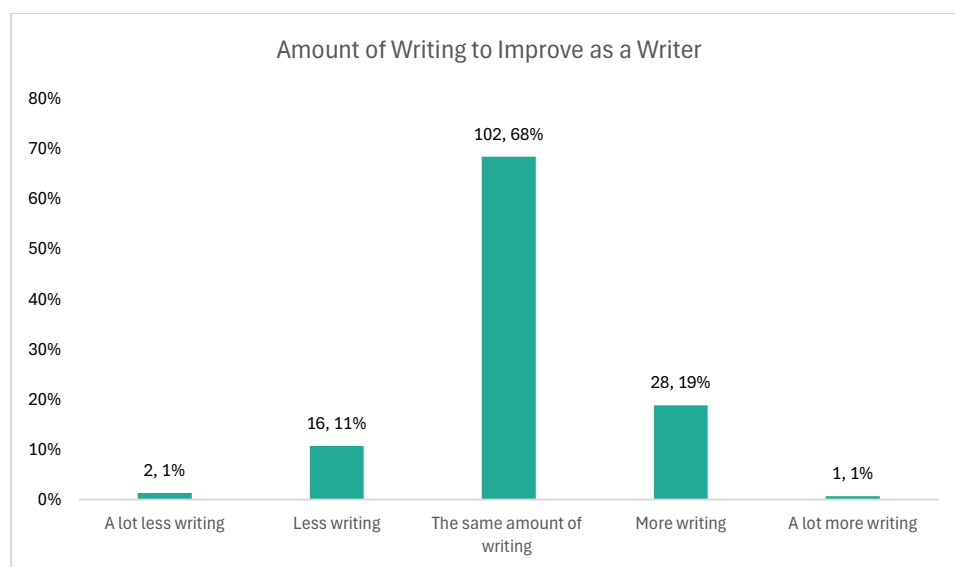


Figure 12. Responses to "In my degree, and to improve my ability as a writer, there should be:" (N = 149)

Table 3. Mean Values of "In my degree, and to improve my ability as a writer, there should be:"

All students		Mean (STD)
Gender (p = 0.066)		3.07 (±0.61)
	F	3.13 (±0.59)
	M	2.85 (±0.67)
	NB/Undecided	3.14 (±0.38)
Academic Year (p = 0.39)		
	1 st year	3.02 (±0.63)
	2 nd year	3.00 (±0.67)
	3 rd year	3.19 (±0.60)
	4 th year+	3.25 (±0.45)
Specialization (p = 0.79)		
	Chemistry	2.96 (±0.60)
	Chemical Biology	3.20 (±0.61)
	Biochemistry	3.08 (±0.62)
	Combined Major	3.04 (±0.64)
Co-op Experience (p = 0.37)		
	Yes	3.20 (±0.56)
	No	3.05 (±0.62)
Primary Language (p = 0.15)		
	English	3.10 (±0.95)
	Other	3.00 (±0.00)
	Multiple primary languages	2.75 (±0.97)
Post Graduate Plans (p = 0.32)		
	Enter the workforce	3.00 (±0.52)
	Professional program	3.08 (±0.53)
	Graduate school (science or non-science)	3.08 (±0.64)
	Undecided	3.07 (±0.73)

As for when the writing should be concentrated within a degree, most students ($n = 100$, 67%) wanted it spread throughout the degree with near even numbers of students answering for earlier in the degree ($n = 21$, 14%) and later in the degree ($n = 28$, 19%). This gaussian distribution is shown in **Figure 13**. There were no statistically significant differences by gender, academic year, major, co-op experience, primary language, or post-graduate plans (**Table 4**).

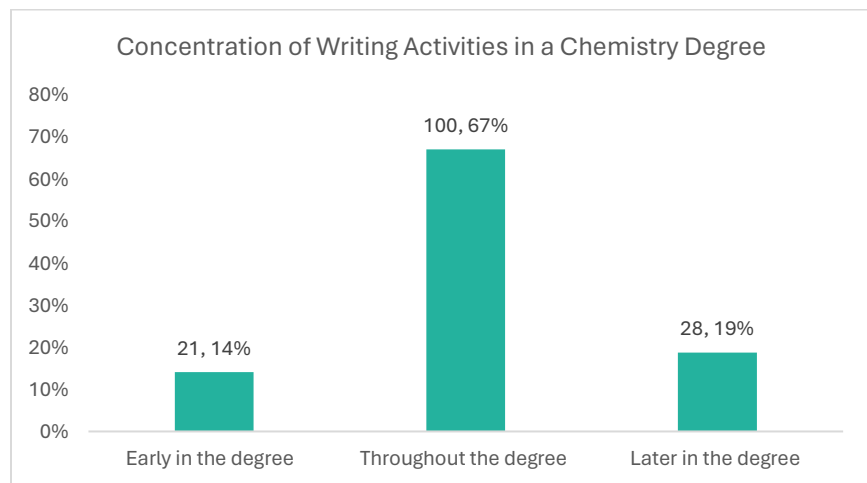


Figure 13. Responses to "When should the majority of writing activities be concentrated in a chemistry degree?" ($N = 149$)

Table 4. Mean Values of "In my degree, and to improve my ability as a writer, there should be:" ($N = 149$)

All students		Mean (STD)
Gender ($p = 0.53$)		2.05 (± 0.58)
	F	2.04 (± 0.54)
	M	2.03 (± 0.64)
	NB/Undecided	2.29 (± 0.76)
Academic Year ($p = 0.27$)		
	1 st year	2.02 (± 0.53)
	2 nd year	2.21 (± 0.63)
	3 rd year	1.90 (± 0.54)
	4 th year+	2.19 (± 0.75)
Specialization ($p = 0.66$)		
	Chemistry	2.06 (± 0.46)
	Chemical Biology	2.09 (± 0.69)
	Biochemistry	2.07 (± 0.57)
	Combined Major	1.89 (± 0.66)
Co-op Experience ($p = 0.57$)		
	Yes	2.13 (± 0.52)
	No	2.04 (± 0.58)
Primary Language ($p = 0.39$)		
	English	2.03 (± 0.59)
	Other	2.27 (± 0.47)
	Multiple primary languages	2.00 (± 0.60)

Post Graduate Plans (p = 0.47)		
	Enter the workforce	2.00 (± 0.58)
	Professional program	1.97 (± 0.67)
	Graduate school (science or non-science)	2.03 (± 0.52)
	Undecided	2.21 (± 0.59)

Students were then asked what types of writing activities should be included within the curriculum of a chemistry degree. Nearly all students (n = 143, 95%) agreed that lab reports should be included, shown prominently in **Figure 14**. There was a large gap between lab reports and the next most selected activity, long writing assignments e.g., essays, research papers and theses (n = 104, 69%), with less but a still significant number of students choosing short answer questions on exams (n = 76, 51%), long answer questions on exam (n = 41, 27%), and creative writing assignments (n = 14, 9%). The low prominence of creative writing could reflect either an availability heuristic, where students don't associate creative writing and chemistry due to the small amount of creative writing opportunities or that students tend towards excluding non-technical writing from their chemistry degree. On average, students chose 3 out of 6 activities, showing that while some activities were preferential over others, many students saw value in including multiple activity types (**Figure 15**). The activities that were most chosen were those commonly found within a chemistry degree: lab reports, research papers, short-answer responses on exams and short writing activities. Less typical activities such as creating writing and essay-style exam questions, common to the arts, were unfavored by the students further supporting the availability heuristic hypothesis.

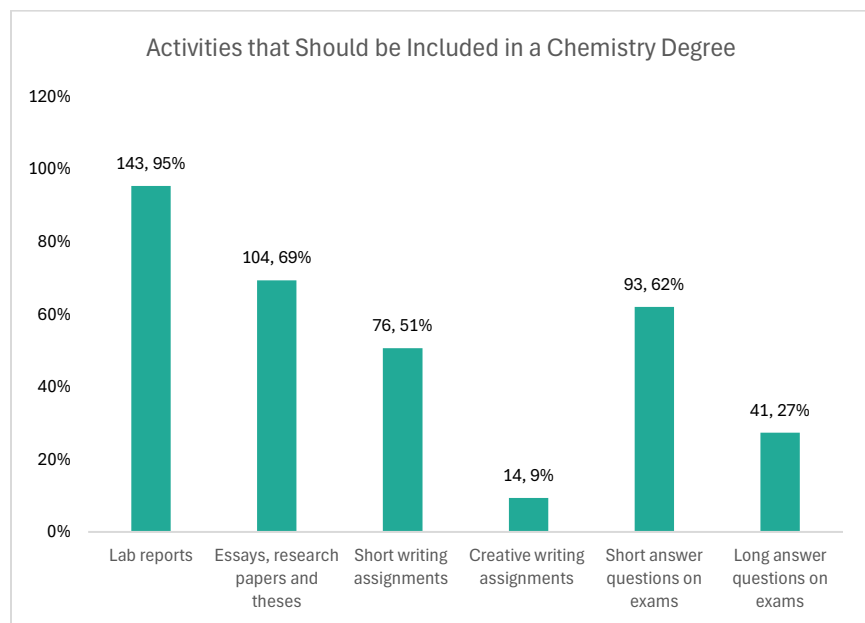


Figure 14. Responses to "What types of writing activities should be in a chemistry degree? Select all." (N = 149, count = 471)

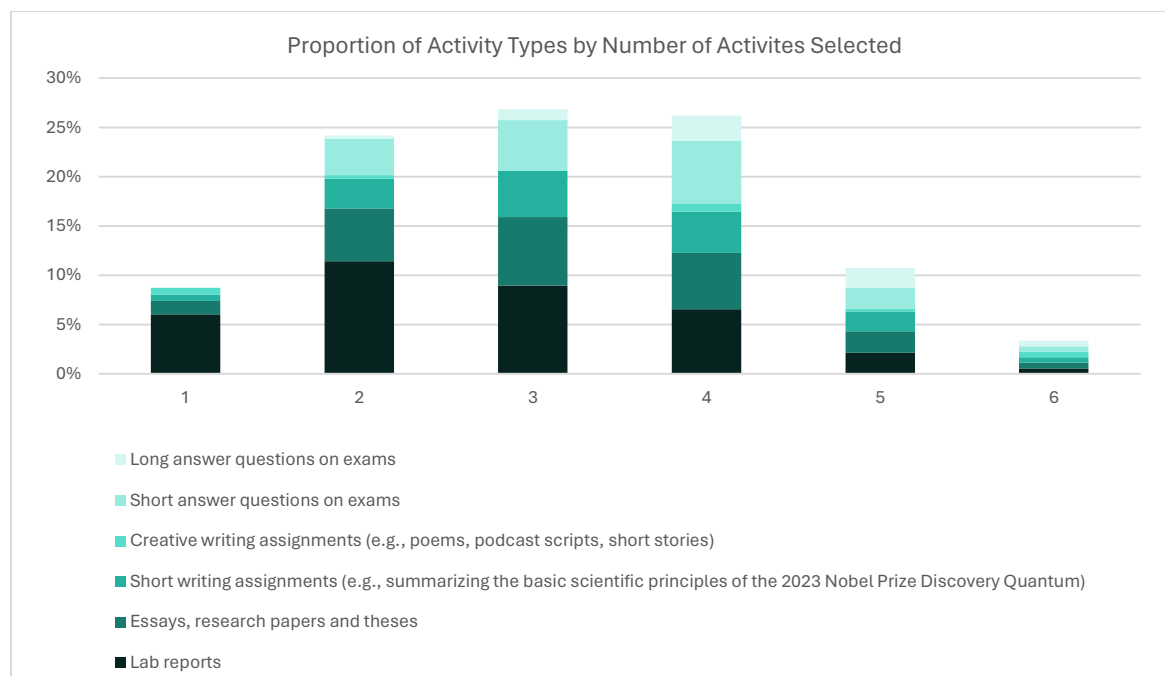


Figure 15. Responses to "What types of writing activities should be in a chemistry degree? Select all." Sorted by number of activity types selected. (N = 149, count = 471)

There were statistically significant differences in agreement of the inclusion of lab reports, research papers, and short writing assignments by year, co-op experience, and primary language (**Table 5**). Proportionally, more first-year students (98%) chose lab reports as an included activity than fourth-year students (81%). Availability of research opportunities in the fourth year could be causing this divide, wherein fourth year students see less value in course-based lab reports after engaging in academic research experience(s). It could also reflect that near the end of their degree, students are fatigued by lab reports and have come to see them in a more negative light. The number of students who selected research papers, theses and essays as an included activity increased with increasing academic year. This suggests that as students progress, they see more value in these types of activities, likely due to a greater exposure to scientific research. Students were split on opinion on short answer on exams with more students in 2nd and 3rd year choosing it than students in 1st and 4th year. This is a very interesting division requiring further investigation to ascertain a potential cause. Lab reports and research papers were also in contention, with students with co-op experience indicating in a greater amount for the inclusion of these activities, than students without. Primary language also influenced how many students included essays and research papers, with fewer students choosing to include it (3:8) if English was not their primary language compared to students for whom English was a primary language (101:37). This is likely due to the difficulties in writing longer assignments for non-native speakers who are not as confident in their English abilities.

Table 5. Chi-squared Test Results of Differences in Demographic Responses to “What types of writing activities should be in a chemistry degree? Select all.” (N = 149)

Subject/Criteria	Gender	Year	Major	Co-op	Primary Language	Post-grad plans
Lab reports	0.36	0.013	0.47	0.041	0.46	0.48
Essays, research papers and theses	0.91	0.0012	0.63	0.047	0.0061	0.51
Short writing assignments	0.11	0.0041	0.33	0.61	0.72	0.86
Creative writing assignments (e.g., poems, podcast scripts, short stories)	0.47	0.75	0.43	0.52	0.066	0.30
Short answer questions on exams	0.31	0.22	0.19	0.32	0.13	0.29
Long answer questions on exams	0.15	0.64	0.30	0.24	0.71	0.47

Note: Statistically significant results are bolded

Having determined what writing activities should be present, students then considered which courses should incorporate writing activities in a chemistry degree. Students answered for inclusion in a variety of course types but overall preferred scientific contexts as shown in **Figure 16**. Chemistry specific writing training courses like CHEM 300 (n = 118, 79%), were the top choice, followed closely by chemistry labs (n = 114, 76%), and scientific writing courses like SCIE 113 (n = 112, 75%). Interestingly, only 83 students (55%) answered that writing activities should be in English writing courses (e.g., WRDS 150, ENGL XXX) and the least popular course type was chemistry lecture courses (n = 38, 8%). The distribution of courses by number of courses selected is shown in **Figure 17**. The most common number of areas where students thought writing should be integrated were three course types and the least common was only one course type. Although a negative response wasn’t offered, the prevalence of students selecting several areas suggests students see value in writing activities in several contexts.

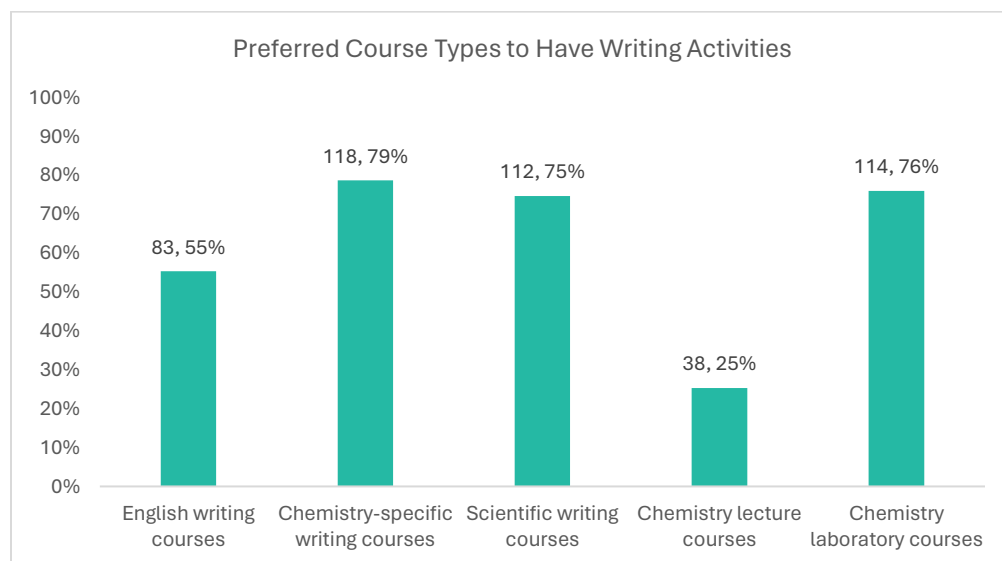


Figure 16. Responses to "Where should writing activities be integrated into a chemistry degree? Select all." (N = 149, count = 465)

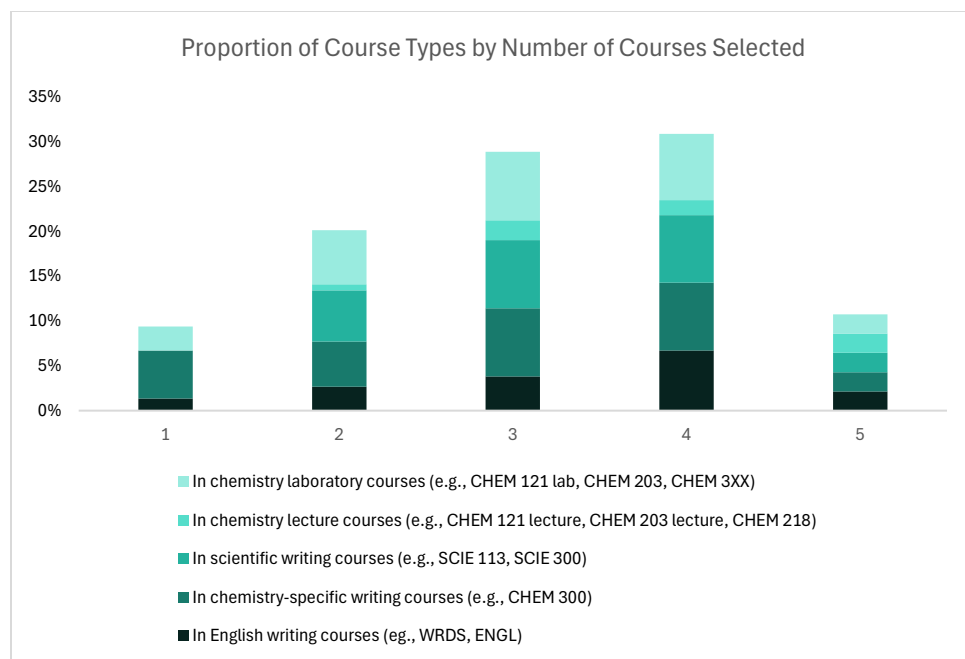


Figure 17. Responses to "Where should writing activities be integrated into a chemistry degree? Select all." Sorted by number of course types selected. (N = 149, count = 465)

There were statistically significant differences in some demographics on whether students chose a particular course area, summarized in **Table 6**. As students' academic year increased, more students selected chemistry specific writing courses, indicating that students who took CHEM 300 saw value in the course. Students who went on co-op were less likely to choose chemistry lecture courses as a place writing should be integrated than those who didn't. However, the sample size for co-op students is small (n = 14, 9%) so careful conclusions must be drawn. Finally, there was a difference by gender for chemistry laboratory courses (p = 0.042). Non-binary, undecided or undeclared students disagreed (3:4) over the inclusion of writing activities within the course and female students, compared to males, agreed with the inclusion of writing activities within labs to a greater degree (81% vs 70%). This conclusion was supported by an adjusted residual analysis.

Table 6. Chi-squared Test Results of Differences in Demographic Responses to "Where should writing activities be integrated into a chemistry degree? Select all." (N = 149)

Subject/Criteria	Gender	Year	Major	Co-op	Primary Language	Post-grad plans
In English writing courses (eg., WRDS, ENGL)	0.10	0.67	0.22	0.11	0.32	0.28
In chemistry-specific writing courses (e.g., CHEM 300)	0.57	0.016	0.11	0.18	0.79	0.55
In scientific writing courses (e.g., SCIE 113, SCIE 300)	0.080	0.93	0.62	0.33	0.078	0.94
In chemistry lecture courses (e.g., CHEM 121 lecture, CHEM 203 lecture, CHEM 218)	0.20	0.67	0.61	0.029	0.13	0.26
In chemistry laboratory courses (e.g., CHEM 121 lab, CHEM 203, CHEM 3XX)	0.042	0.41	0.83	0.21	0.60	0.46

Note: Statistically significant results are bolded

While there are certain courses students felt more strongly for writing being in, most students answered that writing was equally helpful in all chemistry subdisciplines ($n = 87$, 58%). Students were provided several core subdisciplines to choose from (**Figure 18**) and a negative response that writing activities are unhelpful, which only 10 students (7%) chose. When students chose certain subdisciplines as where writing was most helpful, students selected general ($n = 42$, 28%), analytical ($n = 40$, 27%), and organic ($n = 27$, 18%) chemistry the most. Surprisingly, inorganic chemistry received the least number of responses which is interesting given that like general chemistry, inorganic chemistry is conceptually heavy, with many aspects requiring written explanations. Organic chemistry comparatively is primarily mechanistic in language and analytical chemistry, primarily numeric in language. However, this selection could be due to students identifying lab reports as a primary method of science communication and thus selecting courses in which they've written the most lab reports, which for lower years is CHEM 121/123/141, CHEM 203, and CHEM 211.

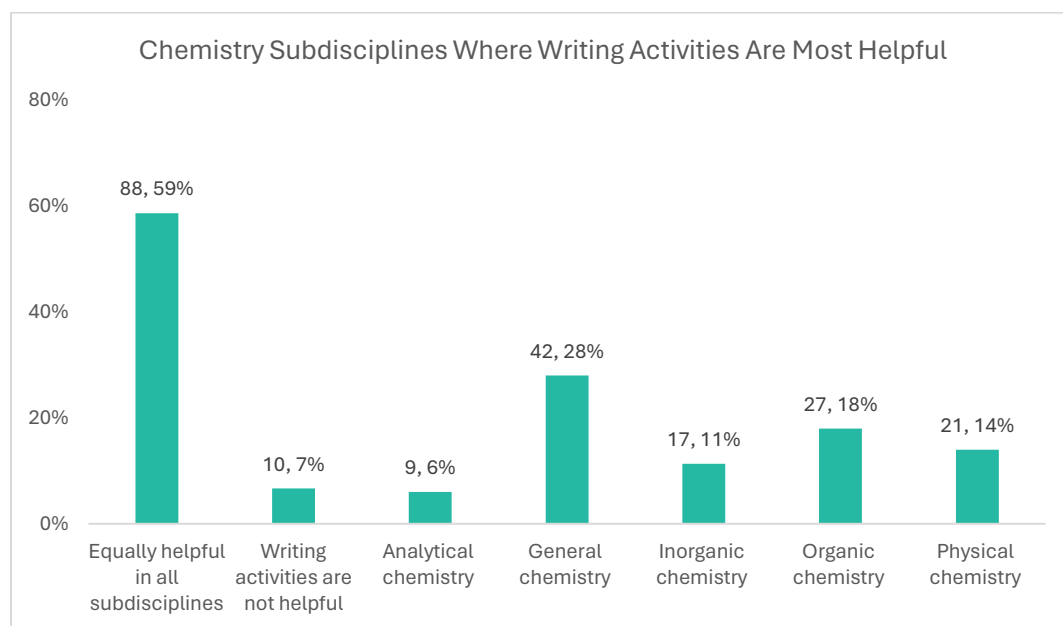


Figure 18. “In what chemistry subdisciplines are writing activities helpful to improve your ability as a writer and/or increase your understanding of the course content? Select all.” ($N = 149$, count = 245)

Students then ranked writing related skills on a scale from not important (1) to very important (4) on importance to (a) their degree (**Figure 19, Table 7**) and (b) their future goals (**Figure 20, Table 8**). Within a chemistry degree, students ranked writing lab reports (3.54 ± 0.66) as the most important skill, followed by clearly communicating ideas to a scientific audience (3.45 ± 0.70). Except for long paragraph or essay questions on a test, most students ranked all other skills as important (3) or very important (4) with mean values above 3. While communication to a general audience (3.05 ± 0.83) was considered lower than to a scientific audience, these questions specifically asked students to consider skills used *within* their degree and should reflect both what students think are important skills and what skills they expect to use. Within the current curricula, students have very

few assignments that address a general audience therefore, communicating to a general audience is less important within their degree than communicating to a scientific audience.

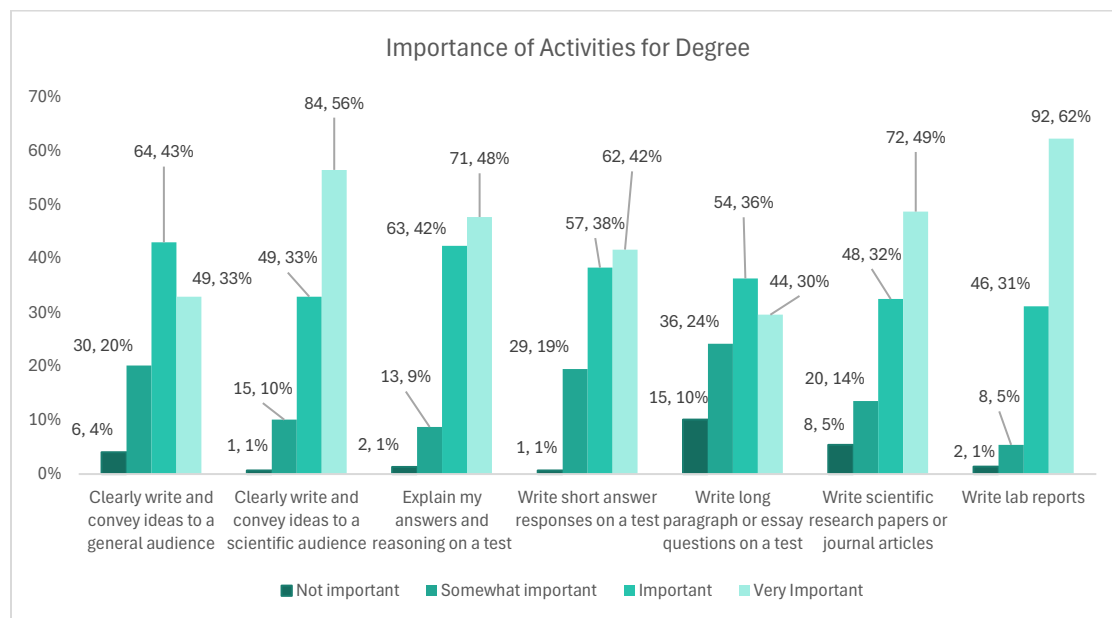


Figure 19. Responses to “How important is it for your degree that you can:” (N = 149)

Table 7. Mean Values of “How important is it for your degree that you can:” (N = 149)

Mean (SD)	Clearly write and convey ideas to a general audience	Clearly write and convey ideas to a scientific audience	Explain my answers and reasoning on a test	Write short answer responses on a test	Write long paragraph or essay questions on a test	Write scientific research papers or journal articles	Write lab reports
All students	3.05 (±0.83)	3.45 (±0.70)	3.36 (±0.70)	3.21 (±0.77)	2.85 (±0.96)	3.24 (±0.89)	3.54 (±0.66)
Gender	(p = 0.55)	(p = 0.59)	(p = 0.84)	(p = 0.11)	(p = 0.0036)	(p = 0.45)	(p = 0.44)
F	3.08 (±0.78)	3.43 (±0.70)	3.38 (±0.68)	3.28 (±0.73)	3.01 (±0.87)	3.25 (±0.86)	3.58 (±0.62)
M	2.91 (±1.01)	3.45 (±0.75)	3.30 (±0.81)	2.97 (±0.85)	2.45 (±1.12)	3.12 (±0.99)	3.41 (±0.75)
NB/Undecided	3.14 (±0.69)	3.71 (±0.49)	3.43 (±0.53)	3.33 (±0.82)	2.29 (±0.95)	3.57 (±0.53)	3.57 (±0.78)
Academic Year	(p = 0.86)	(p = 0.25)	(p = 1.3E-4)	(p = 0.77)	(p = 0.73)	(p = 0.64)	(p = 0.076)
1 st year	3.02 (±0.79)	3.38 (±0.74)	2.83 (±0.96)	3.19 (±0.76)	2.83 (±0.96)	3.17 (±0.9)	3.43 (±0.68)
2 nd year	3.00 (±0.82)	3.47 (±0.61)	3.26 (±0.65)	3.32 (±0.67)	3.00 (±0.88)	3.38 (±0.77)	3.63 (±0.76)
3 rd year	3.19 (±0.98)	3.71 (±0.64)	3.57 (±0.68)	3.10 (±0.89)	2.71 (±1.06)	3.38 (±0.74)	3.71 (±0.56)
4 th year+	3.06 (±0.93)	3.50 (±0.63)	3.63 (±0.62)	3.31 (±0.87)	3.00 (±0.97)	3.31 (±1.07)	3.81 (±0.4)
Specialization	(p = 0.26)	(p = 0.94)	(p = 0.67)	(p = 0.95)	(p = 0.85)	(p = 0.50)	(p = 0.15)
Chemistry	3.04 (±0.79)	3.46 (±0.75)	3.43 (±0.65)	3.26 (±0.77)	2.85 (±0.92)	3.32 (±0.85)	3.44 (±0.69)
Chemical Biology	3.28 (±0.77)	3.50 (±0.67)	3.38 (±0.79)	3.16 (±0.92)	2.94 (±0.95)	3.03 (±1.07)	3.63 (±0.55)
Biochemistry	2.98 (±0.85)	3.40 (±0.69)	3.27 (±0.74)	3.19 (±0.72)	2.77 (±1.02)	3.26 (±0.8)	3.67 (±0.55)
Combined Major	2.84 (±0.96)	3.47 (±0.70)	3.42 (±0.51)	3.21 (±0.71)	2.95 (±0.97)	3.31 (±0.82)	.33 (±0.9)
Co-op Experience	(p = 0.85)	(p = 0.51)	(p = 0.16)	(p = 0.33)	(p = 0.95)	(p = 0.65)	(p = 0.24)
Yes	3.00 (±0.85)	3.33 (±0.72)	3.60 (±0.63)	3.40 (±0.74)	2.87 (±0.99)	3.33 (±0.89)	3.73 (±0.59)

No	3.04 (±0.84)	3.46 (±0.70)	3.33 (±0.70)	3.19 (±0.78)	2.85 (±0.96)	3.22 (±0.88)	3.52 (±0.66)
Primary Language	(p = 0.71)	(p = 0.086)	(p = 0.12)	(p = 0.44)	(p = 0.79)	(p = 0.80)	(p = 0.65)
English	3.07 (±0.83)	3.48 (±0.68)	3.37 (±0.67)	3.21 (±0.76)	2.85 (±0.97)	3.24 (±0.88)	3.55 (±0.65)
Other	2.91 (±0.70)	3.00 (±0.77)	3.00 (±1.00)	3.00 (±1.00)	2.73 (±1.01)	3.09 (±1.04)	3.36 (±0.8)
Multiple primary languages	2.92 (±1.00)	3.50 (±0.80)	3.58 (±0.67)	3.42 (±0.67)	3.00 (±0.85)	3.33 (±0.77)	3.58 (±0.66)
Post Grad Plans	(p = 0.86)	(p = 0.070)	(p = 0.66)	(p = 0.85)	(p = 0.90)	(p = 0.32)	(p = 0.47)
Enter the workforce	2.88 (±0.86)	3.18 (±0.73)	3.29 (±0.77)	3.06 (±0.83)	2.71 (±0.85)	3.35 (±0.7)	3.59 (±0.61)
Professional program	3.05 (±0.94)	3.33 (±0.84)	3.33 (±0.62)	3.23 (±0.71)	2.85 (±0.92)	3.02 (±0.97)	3.40 (±0.77)
Graduate school	3.08 (±0.79)	3.61 (±0.60)	3.44 (±0.66)	3.24 (±0.79)	2.91 (±1.00)	3.33 (±0.88)	3.60 (±0.6)
Undecided	3.07 (±0.78)	3.41 (±0.64)	3.26 (±0.86)	3.19 (±0.83)	2.85 (±1.03)	3.29 (±0.82)	3.59 (±0.63)

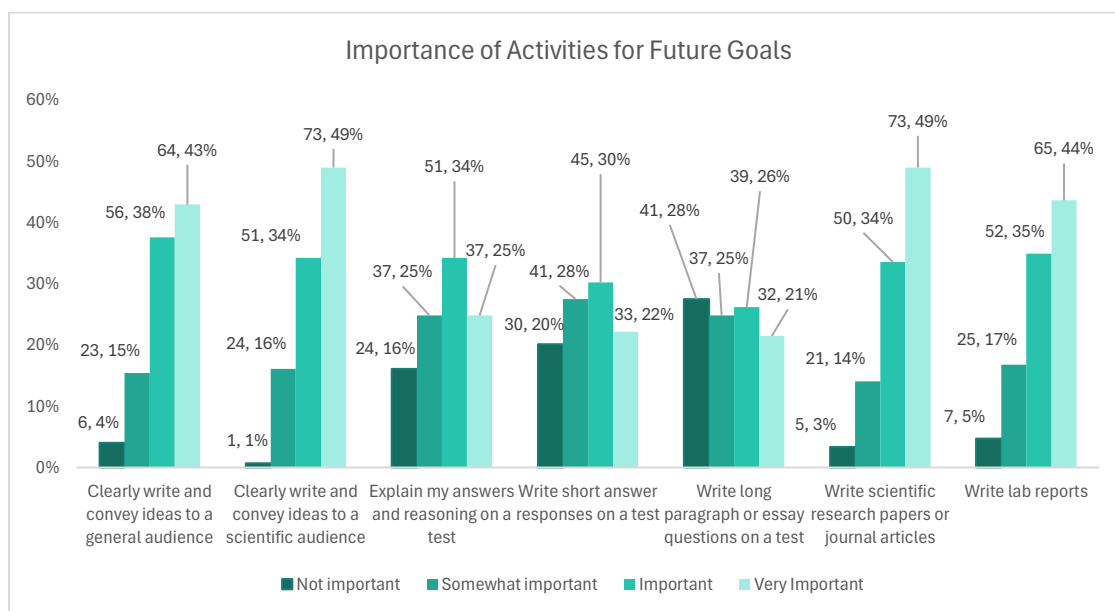


Figure 20. Responses to “How important is it for your future goals that you can:” (N = 149)

Table 8. Mean Values of “How important is it for your future goals that you can:” (N = 149)

M (SD)	Clearly write and convey ideas to a general audience	Clearly write and convey ideas to a scientific audience	Explain my answers and reasoning on a test	Write short answer responses on a test	Write long paragraph or essay questions on a test	Write scientific research papers or journal articles	Write lab reports
All students	3.19 (±0.84)	3.32 (±0.76)	2.68 (±1.02)	2.54 (±1.05)	2.42 (±1.11)	3.28 (±0.83)	3.17 (±0.88)
Academic Year	(p = 0.053)	(p = 0.32)	(p = 0.054)	(p = 0.013)	(p = 0.017)	(p = 0.132)	(p = 0.332)
1 st year	3.14 (±0.83)	3.23 (±0.77)	2.82 (±0.95)	2.73 (±0.98)	2.61 (±1.06)	3.17 (±0.85)	3.10 (±0.90)
2 nd year	3.05 (±0.91)	3.42 (±0.69)	2.11 (±0.94)	1.89 (±0.94)	1.79 (±0.98)	3.32 (±0.89)	3.42 (±0.77)
3 rd year	3.24 (±0.89)	3.48 (±0.75)	2.57 (±1.16)	2.43 (±1.21)	2.19 (±1.25)	3.57 (±0.60)	3.33 (±0.91)
4 th year+	3.63 (±0.72)	3.50 (±0.82)	2.69 (±1.14)	2.38 (±1.09)	2.31 (±1.08)	3.50 (±0.82)	3.13 (±0.81)
Post Grad Plans	(p = 0.30)	(p = 0.038)	(p = 0.79)	(p = 0.24)	(p = 0.36)	(p = 8.6E-5)	(p = 0.045)
Enter the workforce	3.35 (±0.70)	3.12 (±0.78)	2.65 (±1.06)	2.29 (±1.05)	2.29 (±1.10)	3.18 (±0.73)	3.35 (±0.93)

Professional program	3.15 (± 0.84)	3.10 (± 0.85)	2.77 (± 0.93)	2.72 (± 0.92)	2.46 (± 1.05)	2.87 (± 0.95)	2.85 (± 0.99)
Graduate school	3.29 (± 0.82)	3.52 (± 0.66)	2.71 (± 1.03)	2.62 (± 1.05)	2.55 (± 1.15)	3.62 (± 0.6)	3.32 (± 0.77)
Undecided	2.93 (± 0.96)	3.26 (± 0.76)	2.48 (± 1.12)	2.26 (± 1.20)	2.11 (± 1.09)	3.11 (± 0.89)	3.19 (± 0.83)

Considering their future goals, students ranked skills as related to their degree i.e., writing lab reports and test answers as statistically less important and more general skills more important. Communicating to a scientific audience (3.32 ± 0.76) and writing scientific papers (3.28 ± 0.83) became the most important skills for student's future goals, with communicating to a general audience had the largest increase in importance (+0.19) to become the third most important skill. **Table 9** shows the differences in means between the two groups and the T-test results. While not statically significant, the difference between the two means for 'communicating to a general audience' could be students identifying a skill gap between what's necessary for their future goals and the current UBC chemistry curricula and assignments.

Table 9. Difference of Means Between (a) During Degree and (b) After Graduation on Writing Skill Importance

	M (Degree)	M (Future)	Difference of M (Future - degree)	T-test p-value
Clearly write and convey ideas to a general audience	3.05	3.19 (± 0.84)	0.15	0.13
Clearly write and convey ideas to a scientific audience	3.45	3.32 (± 0.76)	-0.13	0.12
Explain my answers and reasoning on a test	3.36	2.68 (± 1.02)	-0.68	9.6E-11
Write short answer responses on a test	3.21	2.54 (± 1.05)	-0.66	1.9E-09
Write long paragraph or essay questions on a test	2.85	2.42 (± 1.11)	-0.44	3.4E-04
Write scientific research papers or journal articles	3.24	3.28 (± 0.83)	0.04	0.70
Write lab reports	3.54	3.17 (± 0.88)	-0.37	6.4E-05

Note: Statistically significant results are bolded

Students then selected the top 1 to 3 skills most useful to them in (a) their degree and (b) after they graduate, with a new set of more technical focused hard writing skills. For their degree, most students ($n = 123$, 83%) chose three skills as the most useful, with 25 students (17%) choosing two, and only 1 student (1%) choosing one skill. This suggests that most students saw many of these skills as useful and applicable within their degree. **Figure 21** shows that the most popular skills were content ($n = 128$, 85%) and writing for scientific audiences ($n = 105$, 70%). Surprisingly, the third most selected skill was ethical behaviour ($n = 66$, 44%) over oral communication ($n = 50$, 33%), mechanics ($n = 32$, 21%) writing for general audiences ($n = 31$, 21%) and creative writing ($n = 8$, 5%). The high selection rate of ethical behaviour reflects the recent increased emphases within the chemical community on ethical practices and the subsequent incorporation of ethical behaviour into education.⁸³ Also surprising, was how few students selected mechanics as one of their top three skills. As such a traditional 'hard' writing skill, one would likely consider mechanics a foundational writing ability, however, most students prioritized other skills. Students likely recognised that scientific communication is often evaluated more critically on content such as argument whereas strict mechanical skill is often evaluated within English courses.

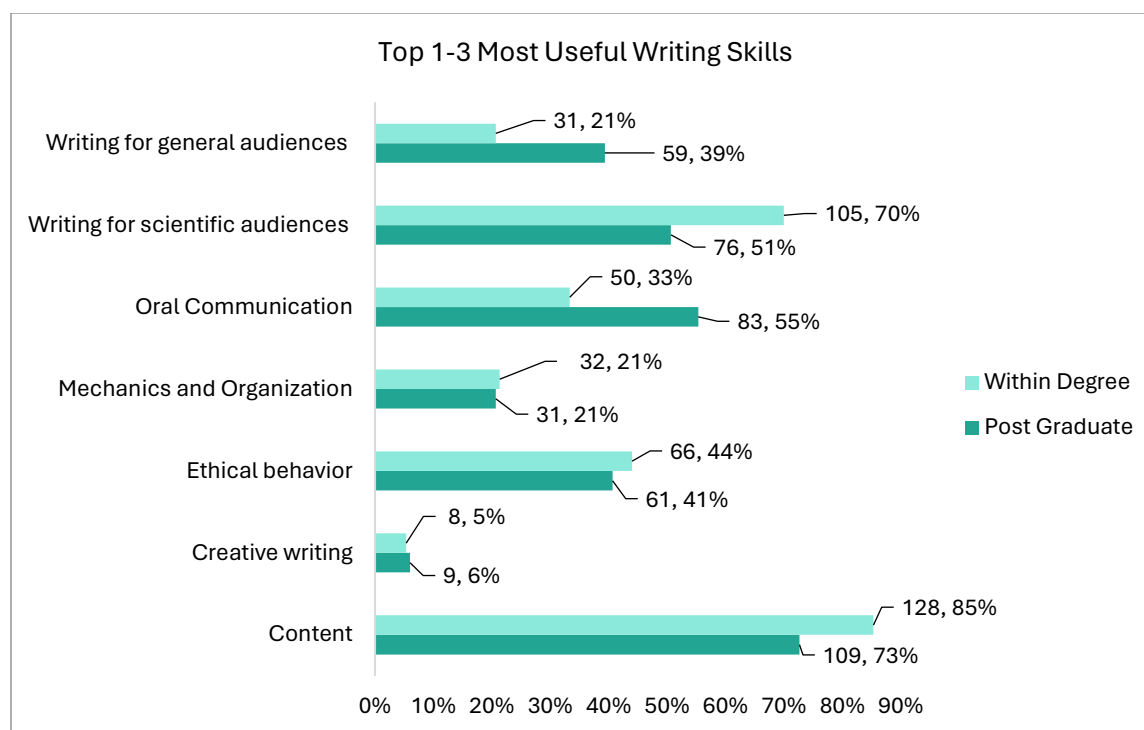


Figure 21. Responses to "Choose the top 1-3 writing related skills that are most useful to you in your:" (a) degree (N = 428, count = 428) and (b) post-graduate plans (N = 149, count = 420)

Students selected oral communication and writing for a general audience at a much higher rate when considering their post-graduate plans. **Table 10** shows the difference in counts between (a) within degree and (b) post-graduate plans. Content remained the most highly chosen skill (n = 109, 73%) followed by oral communication (n = 83, 55%) whilst writing for a scientific audience dropped (n = 76, 51%) as much as writing for a general audience (n = 59, 39%) rose. However, writing for a scientific audience was still the third most selected skill amongst students demonstrating that, although it may be less important for their future, it was still important to most students. The two lowest selected skills – mechanics and creative writing – remained consistent in selections across both groups, suggesting the importance of these skills weren't tied to students' academic or post-graduate goals. The number of skills selected by students diverged from within a degree to after graduation, with increases in students selecting only one of the skills (n = 4, 3%) or three of the skills as most important (n = 134, 90%). This suggests a polarization between students seeing writing related skills en masse as increasingly more or less important to their post-graduate plans.

Table 10. Difference of Means Between (a) During Degree and (b) After graduation for Top 1-3 Writing Skills

	Count (Degree, a)	Count (Future, b)	Difference of M (Future - degree)	T-test p-value
Content (relevance, clarity, originality, coherence)	128	109	-19	0.0063
Creative writing (creativity, worldbuilding, storytelling)	8	9	1	0.80

Ethical behavior (citing, scientific objectivity, information accuracy, etc.)	66	61	-5	0.56
Mechanics and Organization (spelling, grammar, format, structure, cohesion, etc.)	32	31	-1	0.89
Oral Communication	50	83	33	0.00011
Writing for scientific audiences	105	76	-29	0.00054
Writing for general audiences	31	59	28	0.00038

Students were also asked how much they enjoyed a subset of the writing-related activities they ranked for importance. Students overwhelmingly indicated that all types of writing activities were unenjoyable (**Figure 22, Table 11**). Students ranked writing long answers questions on exams (1.44 ± 0.68) as the most unenjoyable activity, with 99 students (67%) ranking it as ‘not enjoyable’ and no students ranking it as ‘very enjoyable’. Conversely, writing short answer questions on exams was the least disliked activity with the highest mean rating (2.06 ± 0.87), however, students still only found the activity somewhat unenjoyable with most still considering it not enjoyable.

Within the demographics, student’s opinions on enjoyability varied based on their primary language. Students for whom English was a primary language had a significantly lower mean enjoyability in writing lab reports than students for whom English was one of their primary languages ($p = 4.3E-4$). Regarding explaining test answers, students for whom English was a co-primary language had significantly higher enjoyment (2.58) than students for whom English was (1.94, $p = 0.018$) or wasn’t (1.81, $p = 0.027$) their primary language. Following the same trend, students for whom English was a primary language rated writing short answers on tests for more favourably than students for whom English is the primary language ($p = 0.011$). This suggests a very interesting trend where students who have multiple primary languages identified several activities as more enjoyable. This could be a result of a small sample size not being representative or could reflect how being fluent in multiple languages changes one’s relationship to communication. Further studies are necessary to see if this trend is a true statistical significance.

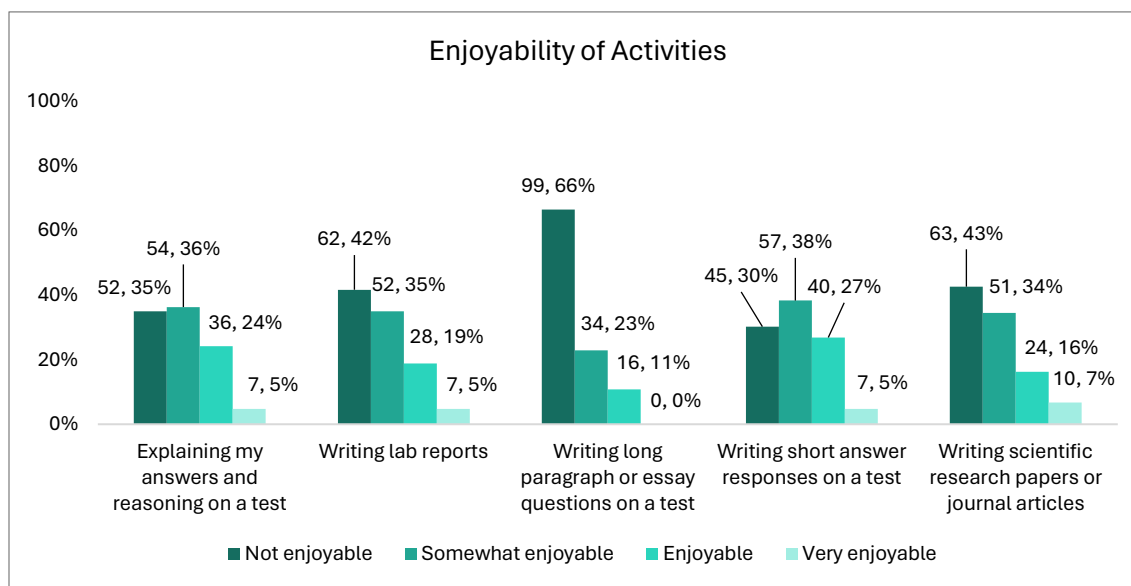


Figure 22. Responses to "How enjoyable are the following activities:" (N = 149)

Table 11. Mean Values of "How enjoyable are the following activities (N = 149)

M (SD)	Explaining my answers and reasoning on a test	Writing lab reports	Writing long paragraph or essay questions on a test	Writing short answer responses on a test	Writing scientific research papers or journal articles
All students	1.99 (±0.89)	1.87 (±0.88)	1.44 (±0.68)	2.06 (±0.87)	1.87 (±0.92)
Gender	(p = 0.19)	(p = 0.51)	(p = 0.89)	(p = 0.23)	(p = 0.88)
F	1.98 (±0.92)	1.92 (±0.87)	1.46 (±0.67)	1.99 (±0.87)	1.88 (±0.94)
M	1.97 (±0.77)	1.73 (±0.91)	1.39 (±0.07)	2.21 (±0.86)	1.82 (±0.88)
NB/Undecided	2.14 (±0.90)	1.71 (±0.95)	1.43 (±0.79)	2.43 (±0.98)	2.00 (±0.82)
Academic Year	(p = 0.12)	(p = 0.34)	(p = 0.62)	(p = 0.014)	(p = 0.91)
1 st year	2.05 (±0.84)	1.91 (±0.89)	1.49 (±0.69)	2.22 (±0.86)	1.84 (±0.87)
2 nd year	1.79 (±0.98)	2.05 (±0.97)	1.32 (±0.67)	1.68 (±0.75)	2.00 (±1.11)
3 rd year	2.14 (±0.96)	1.67 (±0.80)	1.43 (±0.75)	2.05 (±0.97)	1.90 (±0.94)
4 th year+	1.63 (±0.89)	1.63 (±0.81)	1.31 (±0.60)	1.63 (±0.72)	1.88 (±1.02)
Specialization	(p = 0.92)	(p = 0.26)	(p = 0.59)	(p = 0.14)	(p = 0.60)
Chemistry	2.00 (±0.86)	1.96 (±0.91)	1.54 (±0.67)	2.23 (±0.83)	1.98 (±0.96)
Chemical Biology	2.03 (±1.00)	2.03 (±0.90)	1.44 (±0.72)	2.00 (±0.98)	1.84 (±0.99)
Biochemistry	1.91 (±0.81)	1.67 (±0.79)	1.35 (±0.67)	1.85 (±0.73)	1.73 (±0.86)
Combined Major	2.05 (±0.97)	1.79 (±0.98)	1.42 (±0.69)	2.21 (±1.03)	1.95 (±0.85)
Co-op Experience	(p = 0.76)	(p = 0.71)	(p = 0.28)	(p = 0.21)	(p = 0.79)
Yes	1.93 (±0.88)	1.80 (±0.94)	1.27 (±0.59)	1.8 (±0.68)	1.93 (±0.70)
No	2.01 (±0.90)	1.89 (±0.89)	1.47 (±0.69)	2.1 (±0.89)	1.87 (±0.94)
Primary Language	(p = 0.045)	(p = 0.0012)	(p = 0.11)	(p = 0.039)	(p = 0.17)
English	1.94 (±0.89)	1.76 (±0.80)	1.40 (±0.67)	2.00 (±0.86)	1.84 (±0.90)
Other	1.82 (±0.75)	2.18 (±1.08)	1.45 (±0.52)	2.09 (±0.83)	1.70 (±0.82)
Multiple primary languages	2.58 (±0.79)	2.67 (±1.07)	1.83 (±0.83)	2.67 (±0.89)	2.33 (±1.15)
Post Graduate Plans	(p = 0.397)	(p = 0.47)	(p = 0.61)	(p = 0.21)	(p = 0.35)
Enter the workforce	2.06 (±1.09)	1.59 (±1.00)	1.29 (±0.59)	1.88 (±0.86)	1.56 (±0.81)
Professional program	1.87 (±0.83)	1.92 (±0.98)	1.44 (±0.68)	1.97 (±0.84)	1.82 (±0.94)
Graduate school	2.11 (±0.86)	1.94 (±0.87)	1.52 (±0.71)	2.23 (±0.86)	2.00 (±0.93)
Undecided	1.81 (±0.88)	1.78 (±0.64)	1.37 (±0.69)	1.89 (±0.93)	1.81 (±0.92)

Note: Statistically significant results are in bold

Students were also asked about their confidence, very briefly the results shall be discussed insofar, as they provide an aid to assessing how well writing training is progressing within UBC. Students ranged in confidence, feeling more confident in their ability to write short response answers (3.01 ± 0.67) and explain answers on tests (2.87 ± 0.71), than communicating to a general audience (2.72 ± 0.77) or scientific audience (2.45 ± 0.87). Students were also somewhat unconfident in writing lab reports (2.54 ± 0.83), writing long response answers (2.4 ± 0.90), or writing scientific research papers/journal articles (1.94 ± 0.90). **Figure 23** shows the distribution of students' confidence levels. Reported confidence both increased and decreased year-over-year as shown in **Table 12**. Confidence levels of first- and fourth-year students were compared to assess the net confidence change. An overall net gain was seen as a difference between most skills, however, only two of them – writing to a scientific audience ($p = 0.042$) and writing lab reports ($p = 0.023$) – were statistically significant. The absence of a larger confidence gain could suggest that training for these activities is insufficient or if believing that the training is sufficient, could be reflecting a difference in perceived versus actual learning gains.⁸⁴

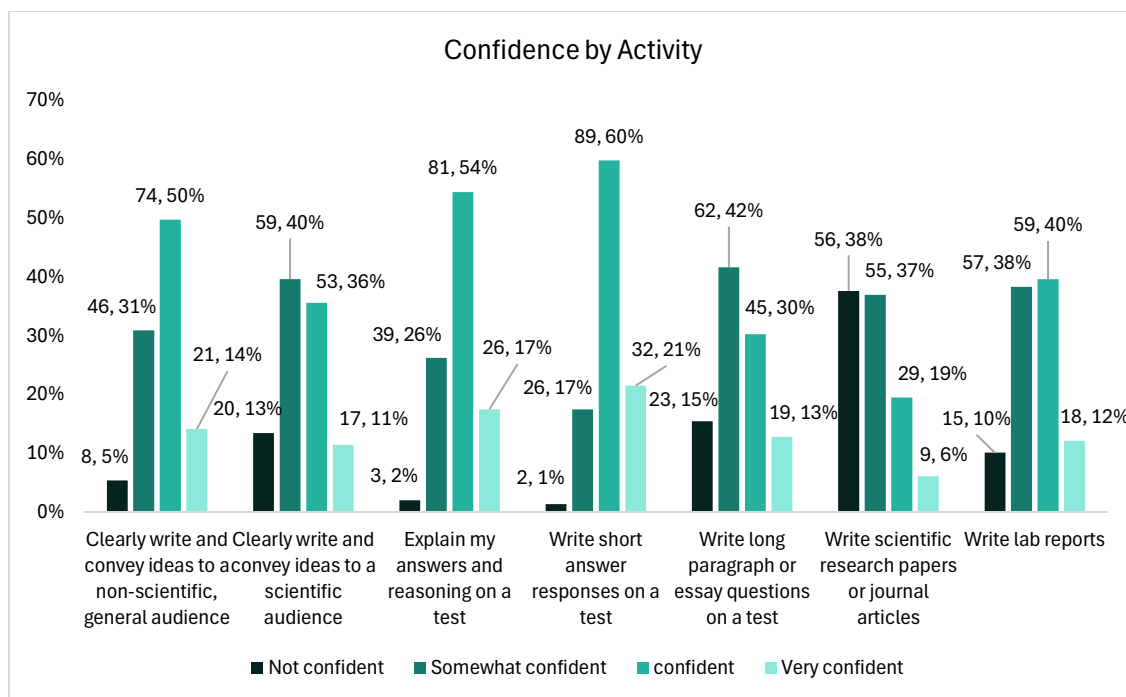


Figure 23. Responses to “How confident are you that you can:” (N = 149)

Table 12. Mean Values of "How confident are you that you can" (N = 149)

M (SD)	Clearly write and convey ideas to a general audience	Clearly write and convey ideas to a scientific audience	Explain my answers and reasoning on a test	Write short answer responses on a test	Write long paragraph or essay questions on a test	Write scientific research papers or journal articles	Write lab reports
All students	2.72 (±0.77)	2.45 (±0.87)	2.87 (±0.71)	3.01 (±0.67)	2.40 (±0.90)	1.94 (±0.90)	2.54 (±0.83)
Academic Year	(P = 0.23)	(P = 0.0072)	(P = 0.031)	(P = 0.21)	(P = 0.029)	(P = 0.24)	(P = 0.051)
1 st year	2.71 (±0.73)	2.34 (±0.85)	2.78 (±0.64)	2.96 (±0.66)	2.27 (±0.86)	1.87 (±0.86)	2.42 (±0.81)
2 nd year	2.58 (±0.77)	2.21 (±0.85)	2.84 (±0.76)	3.05 (±0.62)	2.42 (±0.77)	1.84 (±0.90)	2.68 (±0.75)
3 rd year	2.71 (±0.73)	2.34 (±0.85)	2.78 (±0.64)	2.96 (±0.66)	2.27 (±0.86)	1.87 (±0.86)	2.42 (±0.81)
4 th year+	3.00 (±0.89)	2.81 (±0.75)	2.88 (±0.72)	2.94 (±0.57)	2.63 (±0.96)	2.31 (±1.25)	2.94 (±0.93)
Difference							
4 th - 1 st year	0.29	0.47	0.090	-0.019	0.36	0.44	0.52
T-test p-value							
4 th - 1 st year	0.16	0.042	0.61	0.91	0.14	0.081	0.023

Note: Statistically significant results are in bold

Students were then asked their sentiments regarding their writing proficiency and how writing aided their learning. The questions and results are displayed in **Table 13** and **Figure 24**. Compared to other writing-based skills, students were neutral on writing as less important compared to other content-based skills (3.04 ± 1.10), with a high variance reflecting the spread of answers agreeing and disagreeing with this statement. Students indicated that writing was somewhat helpful to them

in both (a) clarifying thoughts and questions (3.86 ± 0.97) and (b) identifying and fixing misconceptions (3.64 ± 0.97). While similar, these two questions are distinct as they address two different levels of Bloom's taxonomy.⁸⁵ (A) addresses the lower levels of remembering and understanding by clarifying and organizing the information presented. Whereas (b) requires critical analysis and evaluation of the information in a two-step process: first, to identify a misconception, and then reconcile that misconception with other information to rectify it. The mean of (a) was slightly higher than (b) however, the difference was not statistically significant ($p = 0.058$) suggesting students find writing equally helpful across all levels of Bloom's taxonomy. Students also considered themselves as somewhat proficient in writing (3.76 ± 0.87) with most students ($n = 107, 71\%$) agreeing or strongly agreeing.

Table 13. Mean Values of "Rank your agreement with the following statements:" (N =149)

Q#	Question	MEAN	STD
1	I can write well and consider myself proficient in writing.	3.77	0.87
2	(a) Writing helps me clarify my thoughts and questions on a topic.	3.86	0.97
3	(b) Writing helps me identify and fix my misconceptions on a topic.	3.64	0.97
4	Compared to other, content-based skills I learn in my chemistry degree (e.g., synthesis, calibration curves) I think writing is less important.	3.04	1.11
5	I would prefer to be marked on the explanation behind my answer and its logical consistency rather than whether my answer was right or wrong.	3.66	1.14
6	I would be more motivated to write for my job or research than for grades in an academic setting.	3.72	1.08
7	I would produce better writing if my writing was for my job or research than for grades in an academic setting	3.44	1.08
8	I would enjoy writing more for my job or research than for grades in an academic setting.	3.62	1.05

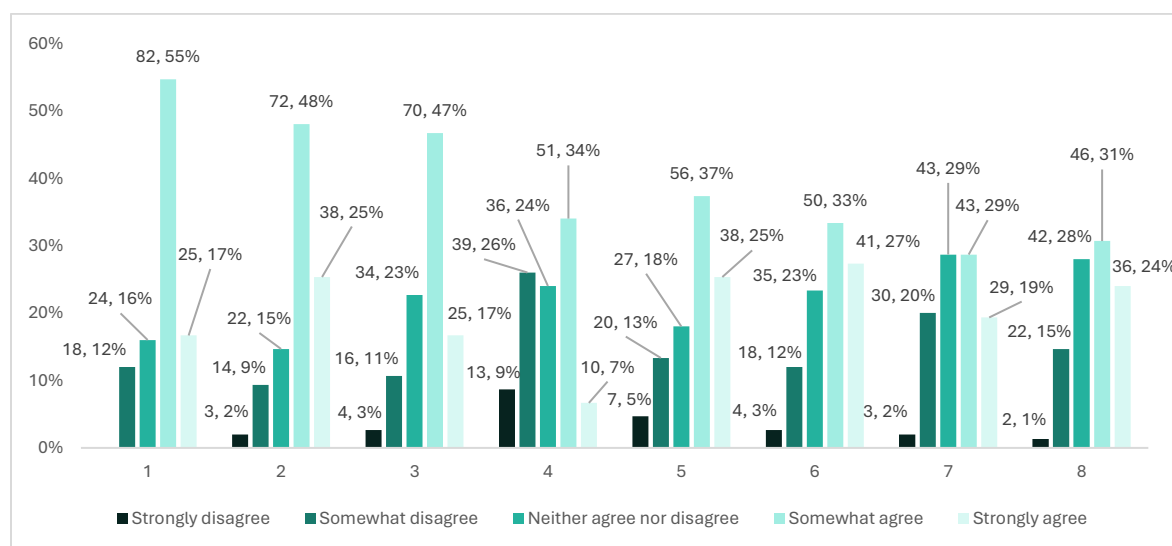


Figure 24. Responses to questions 1 through 8 listed in Table 12. 1 to 4, (N = 149), 5 to 8 (N = 148)

Students were also asked about marking preferences and how their attitudes towards writing would change in a hypothetical situation. Students responded positively (3.66 ± 1.14) to preferring to be marked on the logical consistency of their answers over whether the answer was right or wrong. While not overtly stated, to be marked on their thought process and logic would likely require students to provide a short written, explanation. This question was phrased indirectly to ask students the outcome of incorporating writing rather than the activity to see if students found this method of writing incorporation beneficial without the experiences of other short-response answers biasing their opinion. Placing students in a hypothetical research position or job, students agreed they would have increased motivation to write (6), improved writing performance (7), and greater enjoyability (8), values in **Table 13**. This aligns with other research that shows increased student performance, satisfaction and motivation when course content is aligned to a wider context, justifying the students' efforts.⁸⁶⁻⁸⁸

The final point for discussion of this survey was a series of questions regarding writing training at UBC. These questions were asked about general and scientific writing training as defined in **Table 14**, on a Likert scale of strongly disagree (1) to strongly agree (5). Students responded regarding whether the writing training they received was useful, in sufficient amount to prepare them for their degree or their post graduate plans, if they wanted more writing training, and if they thought universities had a responsibility to provide writing training to students.

Table 14. Definition of the Two Modes of Writing Training Provided to Students in Survey

Term	Definition	Example
General writing training	General, non-specialized writing training that helps with content, grammar, organization (e.g., sentence structure, paragraphs, and flow), and language choice and appropriate use.	ENGL 112
Scientific writing training	targeted, writing training for scientists and scientific contexts. This can include learning to write a scientific research paper, a journal article, sections for a poster, and more.	CHEM 300

On average, students felt neutral about general writing training at UBC (**Figure 25**, top). While students tended towards agreement on whether the amount was sufficient (3.15 ± 1.07), many students still felt it was inadequate for their degree ($n = 45$, 30%) and even more so, their post-graduate plans ($n = 50$, 34%). However, students were split on whether they wanted more ($n = 62$, 42%) or less ($n = 55$, 40%) general writing training. This is an interesting polarization as earlier, when asked whether there should be more or less *writing* within their degree, most students answered for the same amount ($n = 102$, 68%) compared to students who answered the equivalent here 'neither agree nor disagree' ($n = 32$, 21%). This could be reflecting an acquiescence bias, a tendency of students to agree with the given statements or a different opinion. It's possible that students want more writing training without wanting more writing or that being asked whether they feel prepared by the current amount of general writing training caused students to reflect and change their answer. Whilst unsure on the amount, students were pleased with the usefulness of the general writing training provided to them (3.27 ± 1.00), suggesting it is teaching the 'right' skills. When asked if they thought universities *didn't* have a responsibility to provide general writing training most students disagreed (2.00 ± 1.05) concluding that universities should be providing general writing training even within science degrees.



Figure 25. Responses to “Please rank your agreement with the following statements about writing training:” (N = 149). Top: General writing training. Bottom: Scientific writing training.

On scientific writing training, average responses were higher than considering general writing training (**Table 15**). Students reported a higher feeling of preparedness by the current amount of scientific writing training for assignments within their degree (3.49 ± 1.04) and their post-graduate plans (3.14 ± 1.08). However, students still desired more scientific writing training (3.66 ± 1.06) with most students ($n = 91, 61\%$) somewhat or strongly agreeing for wanting *more* scientific writing training. Students also answered that the scientific writing training they received at UBC was useful (3.72 ± 0.93), more useful than their general training ($p = 8.8E-5$). More students also strongly

disagreed (n = 70, 47%) that universities do not have a responsibility to provide scientific writing training than towards general writing training, but the overall difference in means (-0.14) was not statistically significant (0.25). **Figure 25**, bottom shows all responses pertaining to scientific writing training.

Table 15. Difference of Means Between General and Scientific Writing Training

	General writing training M (SD)	Scientific writing training M (SD)	Difference (Sci - Gen)	T-test p-value
The amount of writing training I received at UBC has prepared me for the writing assignments within my degree.	3.15 (± 1.07)	3.49 (± 1.04)	0.34	6.0E-3
The amount of writing training I received at UBC has prepared me for my post graduate plans.	2.89 (± 0.97)	3.14 (± 1.08)	0.25	0.037
I want more writing training in my degree.	3.05 (± 1.21)	3.66 (± 1.06)	0.60	7.4E-6
The writing training I received at UBC was useful.	3.27 (± 1.00)	3.72 (± 0.93)	0.45	8.8E-5
I do not think post-secondary schools (incl. UBC) have a responsibility to provide writing training.	2.00 (± 1.05)	1.86 (± 1.05)	-0.14	0.25

Section 4: Long Answer Thematic Analysis

At the end of the survey, students chose whether they wanted to answer some additional questions. As they were explicitly optional, much fewer students answered these questions (N = 23 to 38, 15 to 26%), however, they still provide valuable data. Question in this section included some personal questions pertaining to students writing practices (omitted in this study) and the free-answer questions. Due to the low volume of responses, we considered responses from all specializations, but clearly indicated which results were in and out-of-scope in **Figures 26 – 29**.

For the first question, students were asked to list any areas they thought more *general* writing training was needed in. **Figure 26** shows that students identified several areas with answers pertaining to specific skills, courses, and periods during a degree. The most common answer included technical skills (n = 7) such as grammar, vocabulary, and writing mechanics – several hard skills. Both in and out-of-scope majors identified this as important. Career-focused items such as cover letters, resumes and emails (n = 5) were also commonly mentioned activities students wanted more support in, as well as being an effective communicator in general (n = 4). When mentioned, students reported wanting more support either earlier in their degree (n = 4) or everywhere across their degree (n=3). The former, aligning with the increased volume of first year respondents (n = 14, 61%) who answered this question.



Figure 26. Responses to "Areas you think more general writing training is needed in." (N = 23, count = 31)

The same question was repeated but in context of scientific writing skills, shown in **Figure 27**. Here, a much broader range of areas were identified with less consensus amongst students. Some students identified lab reports or scientific research papers as specific activities they felt they needed more support in, whereas other students more generally stated they wanted more activities across all science courses. Some students expressed that writing was needed universally, with one student recalling their experience having “peer-reviewed some papers that could’ve been written more coherently by kindergarteners”. Communication reappeared again as a skill both in general and in reference to specific audiences. Hard or technical skills were also identified as important, mimicking the trends seen before between general and scientific writing training.



Figure 27. Responses to "Areas you think more scientific writing training is needed in." (N = 30, count = 47)

After asking what areas needed more training, we asked what writing assignments had been the most useful and why, to which students responded with an answer which constituted all or part of a course, type of activity, and reasoning. Common courses, when mentioned, were the discipline specific writing training courses CHEM 300 and SCIE 113 and chemistry labs. The most common activity was lab reports, followed by scientific research papers, other prominent activities were reviews/summaries and term papers. For why students found these activities helpful, many reasons were supplied. Many students specified lab related skills – with writing lab reports which helped the student learn to “explain [their] ideas thoroughly and concisely” and for another the skill being developed was “[stating] very quantitative observations in a comprehensive way in writing”. Communication skills were prominent as well as increasing understanding of content, general consents of developing scientific literacy were also prevalent, with one student suggesting that the “review article in CHEM 300 ... helped me not only bring together papers and find their commonalities but helped me learn how to read journal articles more efficiently.” Several students also suggested future goals, whether that be the following academic years, research, or other undisclosed plans. What’s evident is that students can, and did, identify many writing related skills and opportunities where they developed on those skills.



Figure 28. Responses to “In your degree, what writing assignments have been the most useful and why?” (N = 38) Top left what courses (count = 35) useful writing assignments were in. Top right Most useful writing assignments (count = 40). Bottom reasons (count = 58) why writing assignments were useful.

While students stated most writing activities were unenjoyable, students were able to identify and recount specific instances of writing activities they enjoyed. The thematic analysis identified the same main categories of ideas, location, activity and reasoning. Some of the same locations were identified, first year labs, discipline specific writing training courses, however, arts courses were also chosen by a few entries. Common activities were lab reports, research papers or essays, and various non-science assignments. The reasons provided were all very interesting. The most cited reason was creative freedom, with students wanting “some kind of agency in what [they’re] writing about” whether that be choosing a topic or “writing essays that disagree with given prompts”. Similarly, students enjoyed when they could write about topics they were interested in. Both topics

are often intertwined, as providing freedom lets students choose topics they are interested in. This positive correlation between student agency and enjoyment is strongly recorded within the literature.^{87,89} Students also enjoyed activities that allowed them to better understand scientific topics or analyse their results. One student highlighted linguistics as being fascinating as “[the assignments], they made me think in a different manner than usual” suggesting that being challenged in their viewpoints can be an enjoyable experience for students. Another student explained that they enjoyed “research essays, because [they] can dive into [their] ideas further and fully articulate them.” Of course, I must add that some students explicitly stated they did not enjoy any writing activities. Some students expressed they had yet to find an enjoyable writing activity “as [they are] still confused on how to go about them” whilst another student bluntly stated that writing assignments were awful.

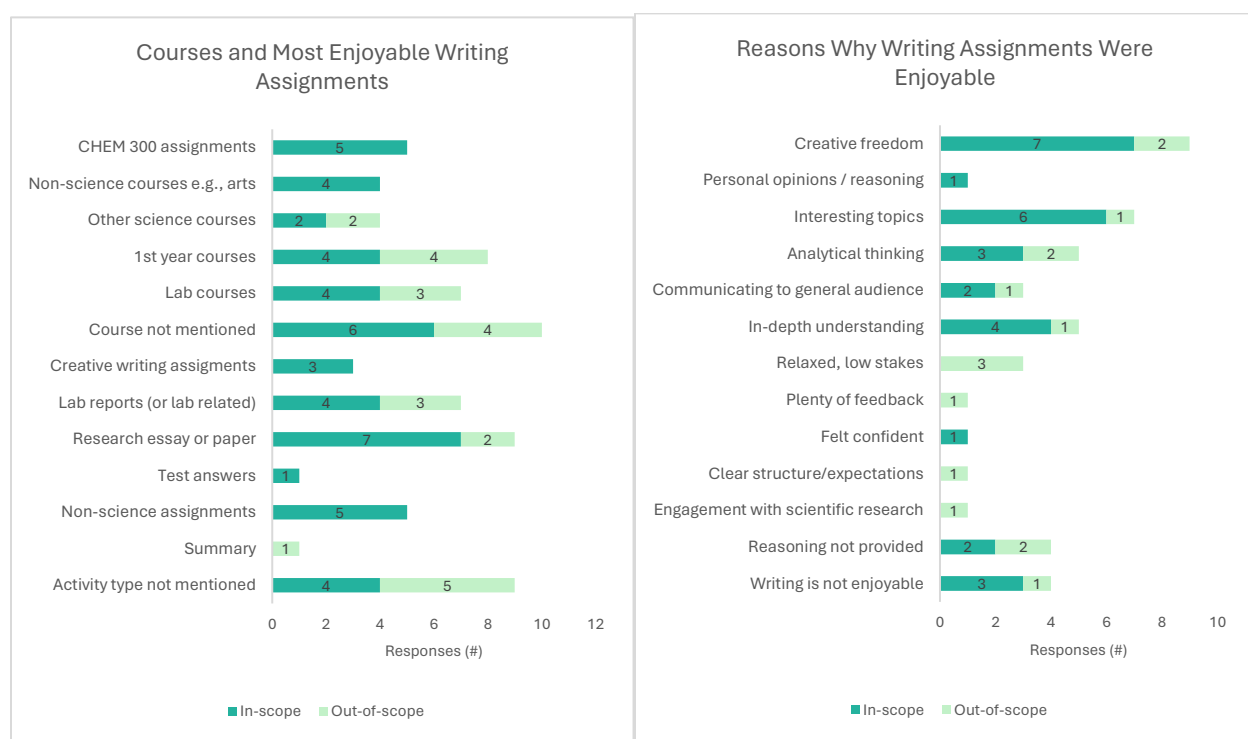


Figure 29. Responses to “In your degree, what writing assignments have been the most enjoyable and why?” (N = 37). Left: what courses (count = 38) the most enjoyable assignments were in, and the assignments (count = 35). Right: reasons (count = 45) why these assignments were enjoyable.

Conclusion

Overall, students had a lot to say about writing and writing training at UBC. **Table 16** summarizes the results of the survey as pertaining to answers to the research themes. The student opinions collected exhibit a wide diversity in student background, experiences, and perspectives. While alternative explanations were offered regarding other causes and considerations for these results in their individual sections, trends become apparent when considered collectively. Three primary trends emerged amongst the data regarding lab reports, exclusion of humanities, and volume of writing and writing training. These trends must be considered cautiously, due to the relatively small sample size and potential for bias amongst the respondents. However, these trends provide some interesting insights into student experience and perceptions despite these limitations.

Table 16. Summary of Survey Results Answering the Five Research Themes

Survey Theme	Survey results and conclusions
(1) Students' understanding of how writing training fits into their chemistry degree, and the role of writing and writing training within chemistry at large	- Students believe universities have a responsibility to provide general and scientific writing training. - Students see scientific writing training as something useful and strongly feel it belongs in their chemistry degree. - Students most strongly associate lab reports as the mode of scientific writing within their degree. - Students identified their top three most useful writing skills as content, communicating to a scientific audience, and ethical behaviour. - Mean importance of all writing related skills (besides essay questions on exams) were at least "important" (3/4).
(2) Students' writing training wants, needs, and satisfaction within their chemistry degree	- Increased skill development on oral communication and communicating to a general audience. - Current general and scientific writing training was useful. - Writing was equally helpful across all chemistry subdisciplines. - All writing activities had a mean value of only somewhat enjoyable (2/4). - o Creative freedom and interesting topics were the most cited reasons for writing assignments that were enjoyable.
(3) Students' relationship with writing and writing training over the course of their degree	- No overall trends - 4th year students felt more confident in writing lab reports and communicating to scientific audiences than 1st year students. - Students in upper years favored inclusion of CHEM 300. - 4th year students indicated lab reports as less important than 1st years.
(4) Impact of different backgrounds and experiences on student perceptions of writing and writing training	- Students whose primary language was not English selected essays as an activity that should be in chemistry less than English speakers. - Students with multiple primary languages identified several activities as more enjoyable than primary English speakers or students with another primary language.
(5) Suggested areas and modes of improvement regarding writing training at UBC	- Increased scientific writing training without increased writing amount. - Varying suggestions for support in specific courses or for lab reports.

Emergent Trends

The first emergent trend was the association of writing and laboratory courses. Students strongly recalled lab courses ($n = 54$, 77%) as areas with high amounts of writing and lab reports ($n = 59$, 84%) as activities performed within these courses. Considering where writing should be integrated, students chose lab courses ($n = 114$, 76%) and for assignments, lab reports ($n = 143$, 95%) topped the list. When asked what subdisciplines writing was most helpful in, the most common options, besides equally helpful, were courses that students would predominantly associate with lab reports at UBC (e.g., organic, general, and analytical chemistry). Writing lab reports was nearly unanimously chosen as an important or very important skill ($n = 138$, 93%) within their degree, and while this importance dropped for post-graduate plans ($n = 117$, 79%), it was still in majority agreement. When asked in what areas students needed more scientific writing training, they freely answered lab reports. When asked what writing activities students enjoyed, they also answered lab reports. It is undeniable that lab reports are essentially interwoven in students' opinions of writing within chemistry.

This prevalent association of writing and labs by students, likely results from experience, reinforced by the symbolism around the discipline. The popular conception of chemistry is as a laboratory science, evident from a simple Google search where the common imagery references lab practices with motifs of beakers, lab glasses, or the lab coat. This symbolism is also a lived experience to students who don these coats to complete a minimum of seven lab courses at UBC within their degree.⁹⁰ Both these experiences and symbolism reinforces lab reports as *the* chemistry writing activity, creating an easy association for students. The ability to write lab reports was also one of only two skills in which fourth years students reported a statistically significantly higher confidence level (2.94 ± 0.93) versus first year students (2.42 ± 0.81). The other skill with a statistically significant improvement was scientific communication, also used in lab reports. This implies that students perceive scientific writing training as primarily concentrated in and effective for writing lab reports.

The second emergent trend was students' exclusion of humanities aligned skills and activities within chemistry. Few students included creative writing activities ($n = 14$, 9%) and long response/essay questions ($n = 41$, 27%) as activities that should be in their degree. Whereas, the other, more typical science communication activities, received an over 50% response rate. Writing long paragraph or essay questions, was the activity students felt least important activity for their degree (2.85 ± 0.96) and their post-graduate plans (2.42 ± 1.11). It was also the most unenjoyable activity (1.44 ± 0.68). However, it wasn't the activity students were least confident in, implying it wasn't perceived negatively because of a skill issue. Mechanics ($n = 32$, 21%, $n = 31$, 21%) and creative writing ($n = 8$, 5%, $n = 9$, 6%) were the least selected skills by students when identifying the one to three most useful writing skills for their degree and post-graduate plans. Collectively, these responses demonstrate that students do not think skills or activities which are typically associated with the arts – creative writing, mechanics, and essay style response on exams, are as important as scientific skills and activities or that they should be incorporated into a chemistry degree. This is well summed up by the difference in selection rate for English writing training courses ($n = 83$, 55%) versus chemistry-specific ($n = 118$, 79%) or scientific ($n = 113$, 75%) writing training courses as areas where writing activities should be integrated in a chemistry degree.

The exclusion of humanities aligned skills and activities could be an experience bias impacting students' values around what should belong within a chemistry degree. The availability heuristic was introduced as a possible explanation for the low inclusion rate of creative writing and long response answers and seems to continue to hold true. Within their degree, students have few opportunities for multi-disciplinary writing, including creative writing. Thus, when considering the role of creative writing, its low assessment causes students to see it as less important and care less about it. The same thought process applies for mechanics, more typically stressed in linguistics or English, and essay style exam questions, more often in arts where students are supposed to leverage an opinion within an argument. The omission of these activities lowers the importance of their skills. Conversely, it is this same process which increased the importance of lab skills demonstrating both effects of the phenomenon.

The narrow association of just scientific writing as important to chemistry is likely harming students enjoyability and dampening their creativity. Students on average disliked all the writing activities proposed, but when asked about activities they did like we learned what made writing assignments enjoyable. The two most popular reasons were creative freedom and an interesting topic, directly suggesting that the things that make writing enjoyable for students are not currently commonly present across their writing activities. The lack of opportunities to practice these skills also results in students not having or valuing creativity as much, even though to be good chemists, students need to be able to creatively apply concepts in real-world applications.⁹¹ Should students be exposed to more creative integrations within chemistry, such as poetry, they'd probably come to like these types of activities and find them beneficial for both WTL and LTW outcomes.^{26,92}

The third emergent trend was regarding the scope of writing and writing training where students appeared to have contradictory opinions. While nearly all students agreed that universities should be teaching both general and scientific writing training, opinions diverged on how and where. When asked directly, most students ($n = 102$, 68%) wanted to maintain the current amount of writing yet later, they indicated wanting more scientific writing training (3.66 ± 1.06). Students indicated that the scientific writing training they received was useful (3.72 ± 0.93) and considered themselves as proficient writers (3.76 ± 0.87) but were only somewhat confident overall (2.56 ± 0.87) in their writing ability regarding various assignments. Students felt that the amount of scientific writing training they received at UBC had prepared them for assignments within their degree (3.49 ± 1.04) but less so for their post-graduate plans (3.14 ± 1.08), with a statistically significant mean difference of 0.36 ($p = 0.0039$).

These seemingly divergent opinions are because students are assessing themselves in different areas and to different standards. Proficiency in writing encompasses all skills but only requires students to consider themselves as a decent writer, which is a tendency many students will have. This positive association is further encouraged by the question which has a positive lean, possibly resulting in an acquiescence bias.⁹³ Whereas asking directly about their confidence in certain skills challenged students' self-beliefs and they may have estimated conservatively, causing divergent results. Considering lab reports as the predominant scientific writing, if students are taught how to write lab reports and feel prepared to write lab reports, they will therefore see scientific writing training as useful and preparing them for the assignments within their degree, even if it is missing and neglecting other skills. Other skills, that students may identify as important for their future thus addressing the lower level of perceived preparedness from the current scientific writing training at UBC. Students may also be differentiating being prepared from being confident. For example, even

with a lot of studying students can still feel nervous and unconfident when going into an exam. Similarly, wanting more writing training and wanting the same amount of writing aren't opposing ideas. Students can want to maintain the same volume of writing but want training that develops skills currently overlooked within their degree but important to their post-graduate plans, such as communicating to general audiences and oral communication.

Reworking current writing activities such as lab reports could offer increased writing training without burdening students with more writing. Students currently write many lab reports across at least seven lab courses in their degree. While these reports are useful for content, first- and second-year lab reports do not offer a lot of value in scientific writing training and could be adapted into something more beneficial to students. In 2001, Tilstra⁹⁴ reworked an entry-level laboratory course curriculum with the goals of familiarizing students with journal articles and providing scientific writing training. Four of the fifteen experiments were chosen to be used as teaching opportunities, with students provided an explanation and handout detailing what skill they were expected to develop and demonstrate within that week's report. The four reports started simply but became more complex, allowing students to build on the skills initially introduced. Alongside this, students chose and explored a recent journal article, drawing parallels between what they were doing in their experiments and where and how similar types of information were presented within their article. Students liked this approach, citing increased clarity in expectations and instruction as well as seeming to show improved performance in upper-year lab courses. While the feasibility of a similar program at UBC is unknown, this demonstrates that it would be possible to change the types of writing already present to provide better writing training without substantially increasing the volume of writing.

Integrating creative writing may be more difficult and require an increased amount of writing. To ease this burden, students could be provided with low-stakes assignments that offer creative writing practice as a WTL and LTW tool under easy to meet criteria. This would allow students to practice writing and receive feedback in a low-stress, more enjoyable environment.⁶⁴ Whilst not in scope, a few students mentioned liking LFS 150 for exactly this reason, "because there are lots and lots of feedbacks [sic] about our writing assignments without any consequences."

Literature Comparison

In the introduction and methodology, we stated that some questions were designed similar to the SSSI to offer a value of comparison. While the parallels are not exact, as this survey asked about more specific writing activities and skills, we can compare the average values of students' responses across all writing skills with the importance, future use, and confidence reported in the literature. **Table 17** shows that this survey reported mean results similar to those reported by Varsavsky et al.⁴⁸ and Mercer-Mapstone and Matthews^{50,51}. While some variations in means were present between the results reported here and the other surveys, they were all within the standard deviation. Confidence regarding the skills was lower than the values reported by Varsavsky et al. and Mercer-Mapstone and Matthews, however, when considering a more inclusive metric, writing proficiency, the results were very closely aligned. No statistical analysis was performed between these results due to the categories being slightly different. Overall, the alignment with values reported in the literature suggests UBC is providing writing training comparable to other universities. This indicates that the results of this survey may be generalizable beyond UBC.

Table 17. Comparison of Survey Outcomes to Literature Values

M SD [Range]	Importance (within degree)	Importance (post- graduate plans) / future use	Confidence	Proficiency in writing
This survey	3.24 (+0.82) [2.85 – 3.54]	2.94 (+1.00) [2.42 – 3.32]	2.56 (+0.87) [1.94 – 3.01]	3.02 (±0.70) ^a
Varsavsky et al. ⁴⁸	3.70 (+0.51)	3.07 (+0.87)	3.24 (+0.69)	-
Mercer-Mapstone and Matthews (2017) ⁵¹	3.51 (±0.59) ^β	3.00 (±0.84) ^β	3.00 (±0.63) ^{αβ}	-
Matthews and Mercer- Mapstone (2018) ⁵⁰	3.47 (±0.61)	3.00 (±0.87)	-	-

^a normalized to a 4-point scale

^β pertaining specifically to scientific writing

Future work

Future work on this project includes further data analysis, successive survey iterations and holding focus groups. As an exploratory study, this survey covered many questions and themes. While an overview was provided here, there are more analyses to be performed on the data. For example, no tracing of students answers to find commonalities was performed, such as seeing if students who reported low confidence in writing lab reports also had similar answers to other questions. Repetitions of the survey in future years will strengthen the data by providing a greater sample size which will aid in determining if the emergent trends were actual trends or just a result of bias from the study. This survey occurred at an opportune time as with future iterations it can trace how student opinions change as SCIE 113 becomes required for all incoming science students and students' progress through their degree with this scientific writing training. Focus groups were planned, but not able to be held before the completion of this report. Nineteen 3rd and 4th year students indicated interest in participating and were emailed by the survey team in acknowledgement. Now that the survey data has been parsed, focus group questions can begin to be developed, invitations sent out, and focus groups held to follow-up on the interesting trends, oddities and collective opinions presented within this report.

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Appendix

Appendix A – Supporting Materials for Each Section

Section 1: Supporting Materials for Introduction

Table 18. Tabulated responses of Graduate students' writing training experience, reproduced from Ritchie et al., 2022¹⁴

Skill	% No training	% In grad institute	% Outside grad institute	% N/A
A: identify and understand a suitable target audience	46	24	28	2
B: use language that is appropriate for my target audience	28	33	37	3
C: identify the purpose and intended outcome of a science communication project	41	26	29	4
D: consider the levels of prior knowledge in my target audience	24	34	38	4
E: separate essential from non-essential factual content in a context that is relevant to the target audience	39	32	26	3
F: use a suitable mode and platform to communicate with the target audience	53	21	24	3
G: consider the social, political, and cultural context of the scientific information	50	18	28	4
H: understand the underlying theories leading to the development of science communication and why it is important	58	16	21	5
I: promote audience engagement with the science	44	24	28	4
J: use the tools of storytelling and narrative	38	24	34	4
K: encourage a two-way dialogue with the audience	44	21	29	6

Note: Colour coded based on response (%): <5%, 10-20%, 20-30%, 30-40%, 40-50%, >50%

Section 2: Supporting Materials for Demographics

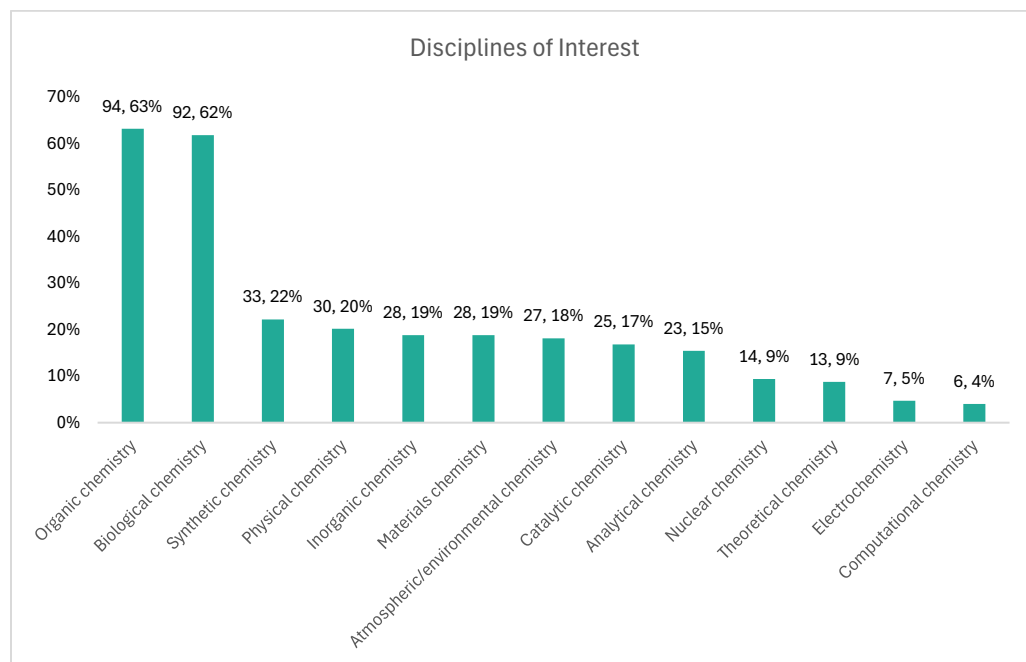


Figure 30. "Please select the 1-3 subdiscipline(s) of chemistry that you are most interested in:" (N =149, count = 420)

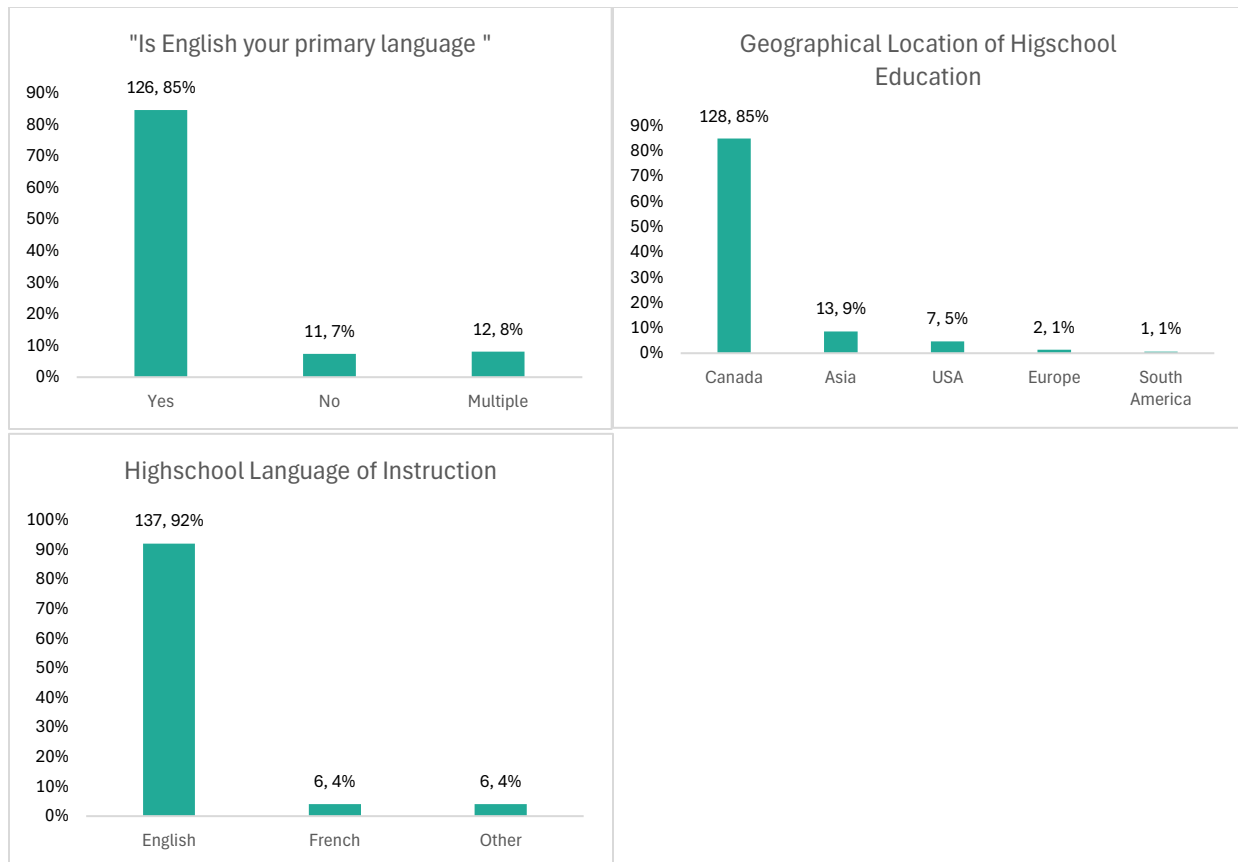


Figure 31. Top, left: "Is English your primary language? I.e., the language you speak most often, feel most comfortable using, perform everyday activities using, think in, etc." (N = 149). Top, right: "In which geographical area(s) did you complete your high school education (years 10, 11, and 12 or their equivalent).” (N = 149). Bottom: "What was the primary language of instruction during your high school education?" (N = 149).

Section 3: Supporting Materials for Student Experiences

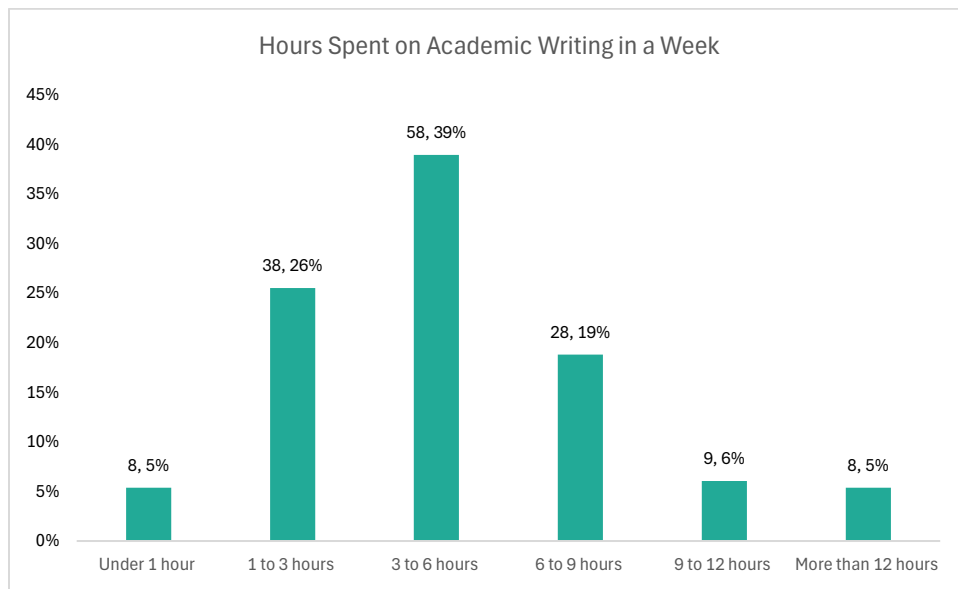


Figure 32. Responses to "How many hours do you spend in an average week on academic-related writing activities? (E.g., writing, assignments, papers, notes, etc.)?" (N = 149)

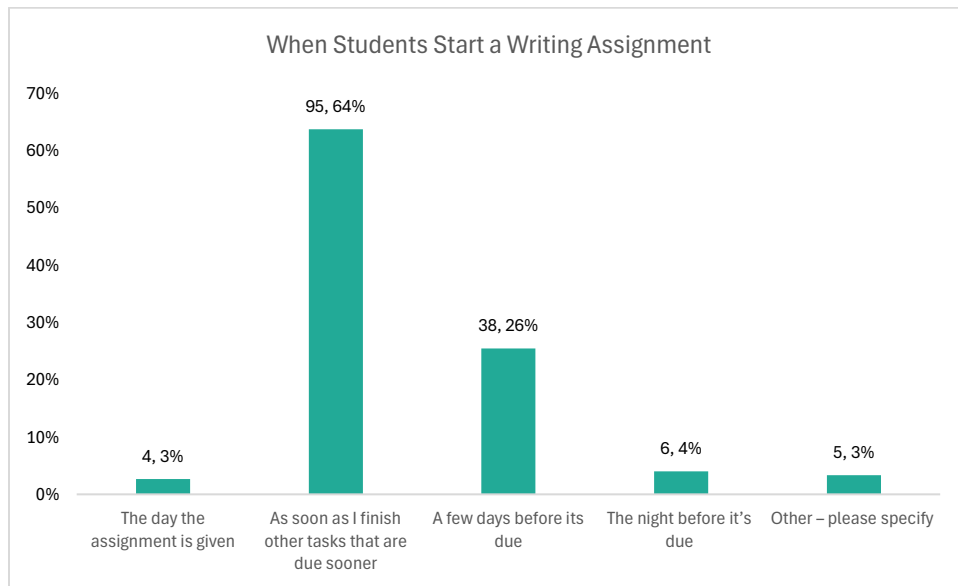


Figure 33. Responses to "When, on average, do you start a writing assignment (i.e., essays, research papers, reports, and stories)?" (N = 149). Other includes week or weekend before it's due, day it's due, and several weeks in advance.

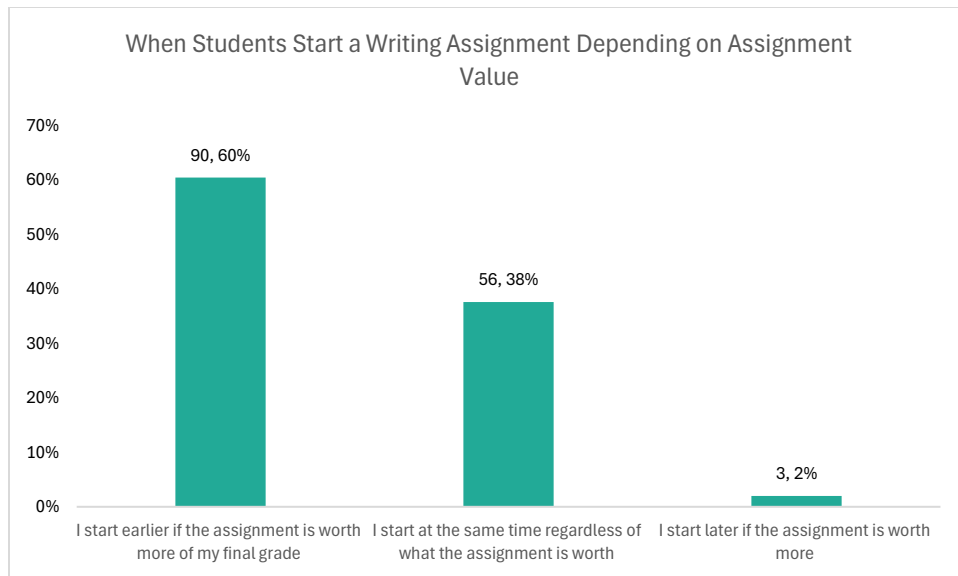


Figure 34. Responses to "How does the amount a writing assignment is worth towards your final grade change when you start that writing assignment?" (N = 149)

Appendix B – BC Student Outcome Survey – UBCV Results

Appendix C – BREB Application Form

Appendix D – Survey

Appendix E – Attached Statistical Tests from Statistics Kingdom (ZIP file)